



US012402349B2

(12) **United States Patent**
Guha et al.

(10) **Patent No.:** **US 12,402,349 B2**
(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **GATE-ALL-AROUND INTEGRATED CIRCUIT STRUCTURES HAVING DEVICES WITH CHANNEL-TO-SUBSTRATE ELECTRICAL CONTACT**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 147 days.
This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/379,554**

(22) Filed: **Oct. 12, 2023**

(65) **Prior Publication Data**
US 2024/0038889 A1 Feb. 1, 2024

Related U.S. Application Data

(63) Continuation of application No. 16/719,222, filed on Dec. 18, 2019, now Pat. No. 11,824,116.

(51) **Int. Cl.**
H10D 30/69 (2025.01)
H10D 30/60 (2025.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 30/611** (2025.01); **H10D 30/62** (2025.01); **H10D 30/6219** (2025.01); **H10D 30/6735** (2025.01); **H10D 62/119** (2025.01)

(58) **Field of Classification Search**
CPC H01L 21/02532; H01L 21/20; H01L 21/823475; H01L 21/823431;
(Continued)

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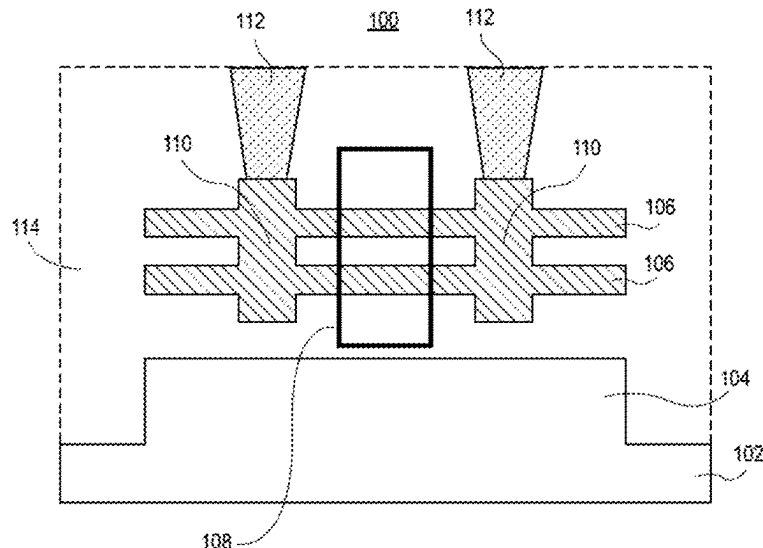
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(57) **ABSTRACT**

Gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact are described. For example, an integrated circuit structure includes a first vertical arrangement of horizontal nanowires above a first fin. A channel region of the first vertical arrangement of horizontal nanowires is electrically coupled to the first fin by a semiconductor material layer directly between the first vertical arrangement of horizontal nanowires and the first fin. A first gate stack is over the first vertical arrangement of horizontal nanowires. A second vertical arrangement of horizontal nanowires is above a second fin. A channel region of the second vertical arrangement of horizontal nanowires is electrically isolated from the second fin. A second gate stack is over the second vertical arrangement of horizontal nanowires.

20 Claims, 18 Drawing Sheets



- (51) **Int. Cl.** H10D 64/251; H10D 62/134; H10D 62/137; H10D 62/184; H10D 62/121; H10D 84/401; H10D 84/811; H10D 84/013; H10D 84/0149; H10D 84/038; H10D 84/83; H10D 86/441; H10D 86/60; H10D 84/0128; H10D 84/0135; H10D 84/0144
- H10D 30/62** (2025.01)
- H10D 30/67** (2025.01)
- H10D 62/10** (2025.01)
- H10D 64/01** (2025.01)
- (58) **Field of Classification Search**
- CPC H01L 21/823821; H01L 21/845; H01L 21/823418-823425; H01L 21/823814; H01L 23/52; H01L 27/088; H01L 27/0886; H01L 27/0924; H01L 27/1211; H01L 29/0673; H01L 29/0669; H01L 29/41791; H01L 29/41725; H01L 29/42392; H01L 29/66545; H01L 29/7831; H01L 29/785; H01L 29/7855; H01L 29/41783; H01L 29/0843-0891; H01L 29/66636-66643; H01L 29/66795-66818; H01L 29/7856; H01L 29/04-045; H01L 29/16-1608; H01L 29/18-185; H01L 29/22-2206; H01L 29/36-365; H01L 2029/7857-7858; H01L 2924/13067; Y10S 977/938; H10B 12/36; H10B 12/056; H10D 30/611; H10D 30/62; H10D 30/6219; H10D 30/6735; H10D 62/119; H10D 30/43; H10D 30/6757; H10D 10/60; H10D 30/797; H10D 64/017; H10D 30/014; H10D 62/371; H10D 62/834; H10D 64/01;
- See application file for complete search history.
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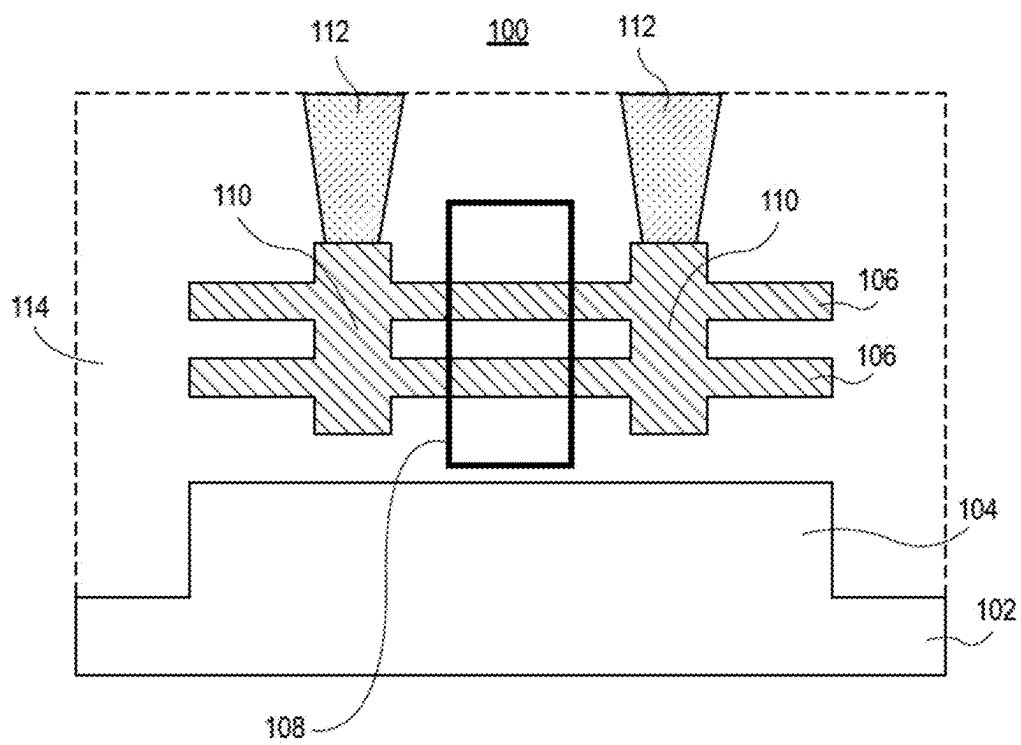


FIG. 1

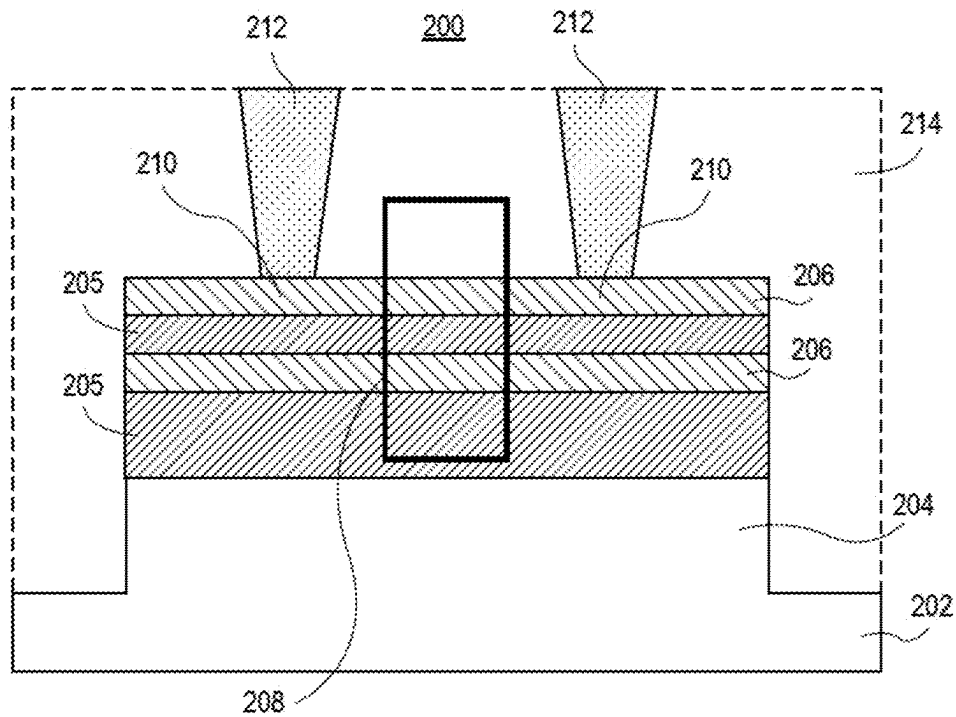


FIG. 2A

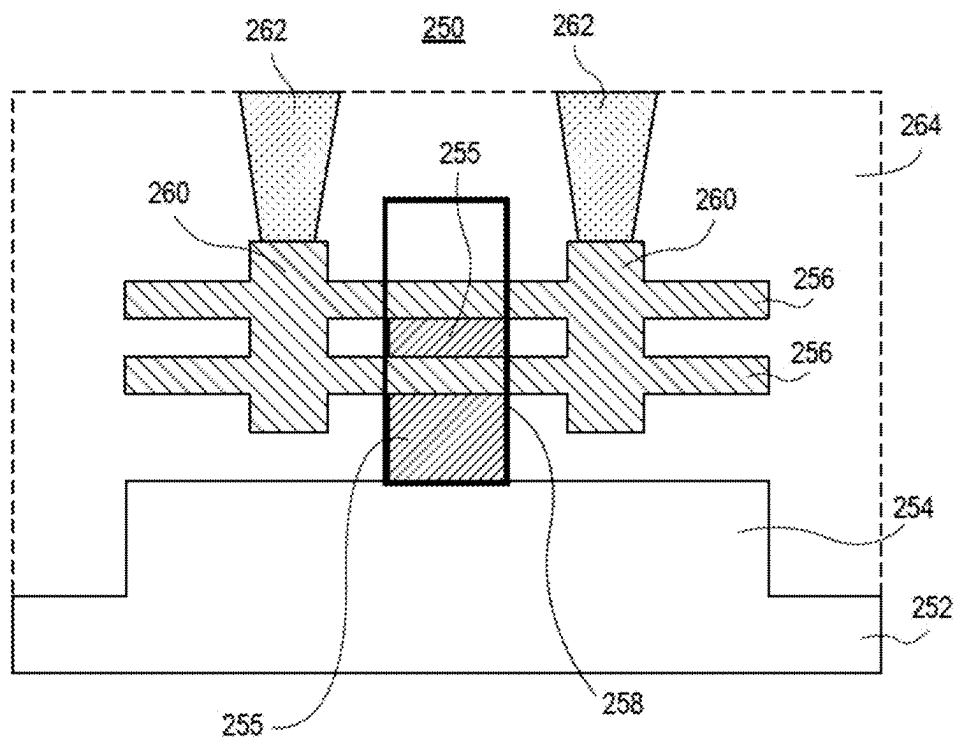


FIG. 2B

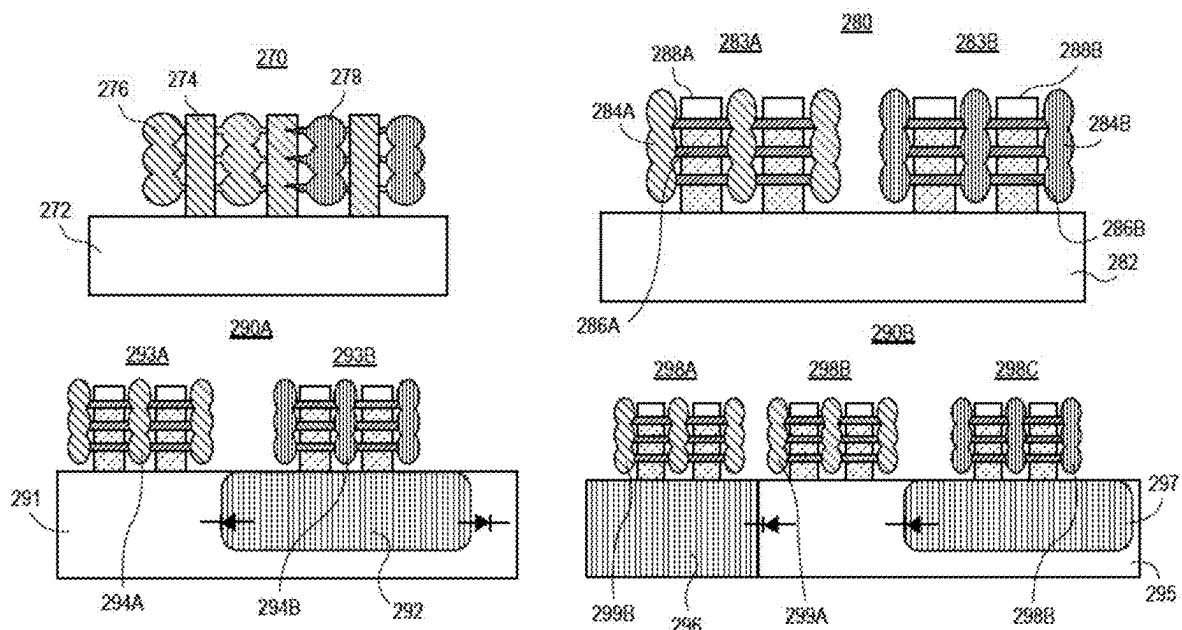


FIG. 2C

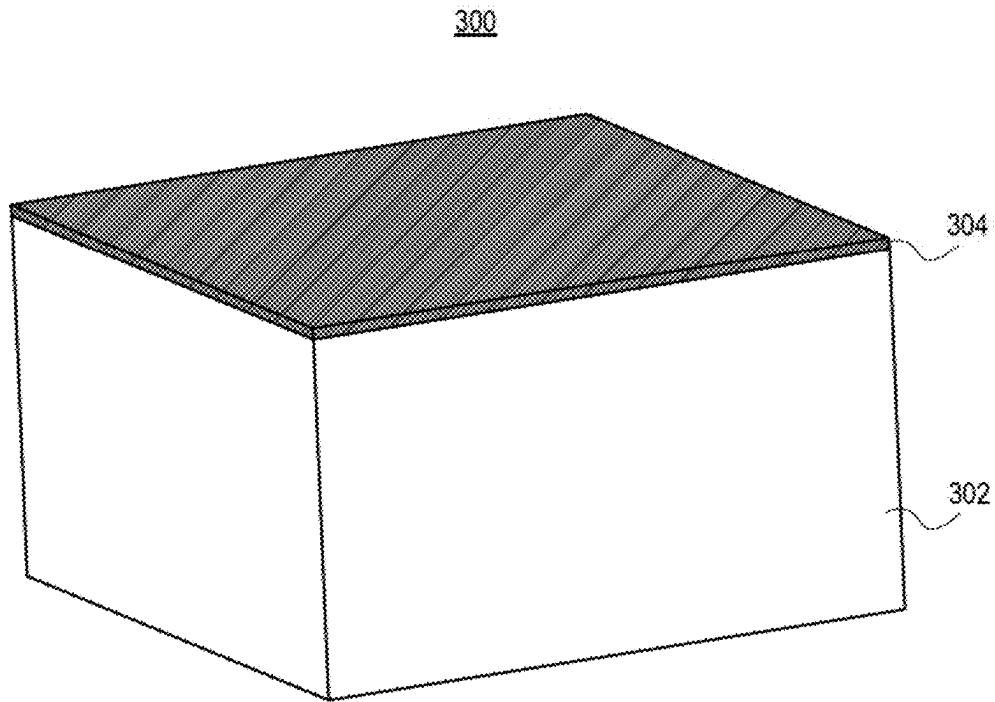


FIG. 3A

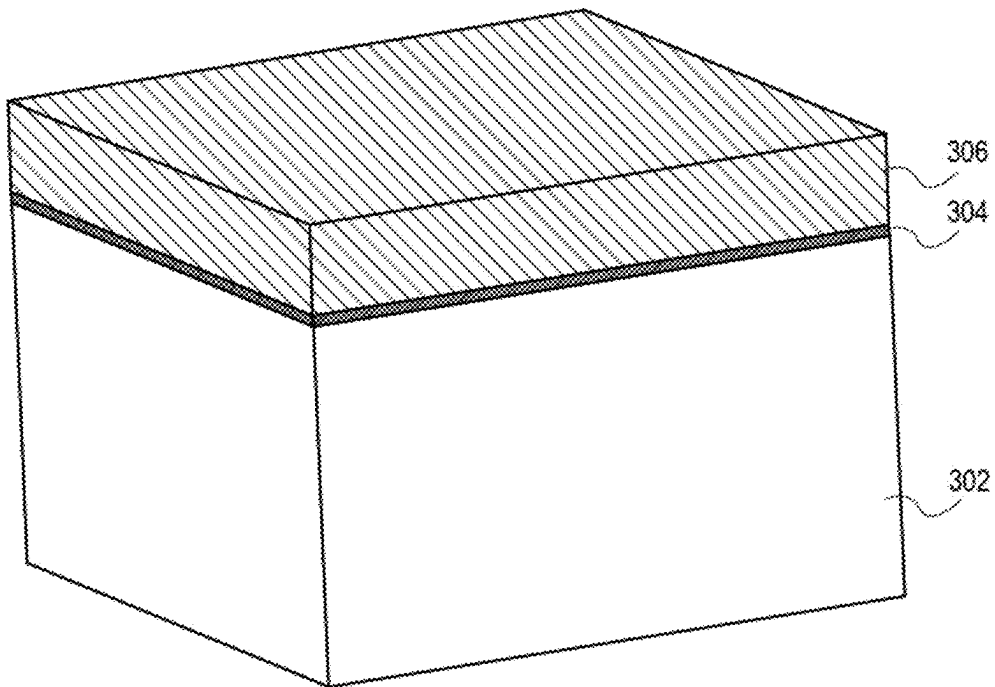
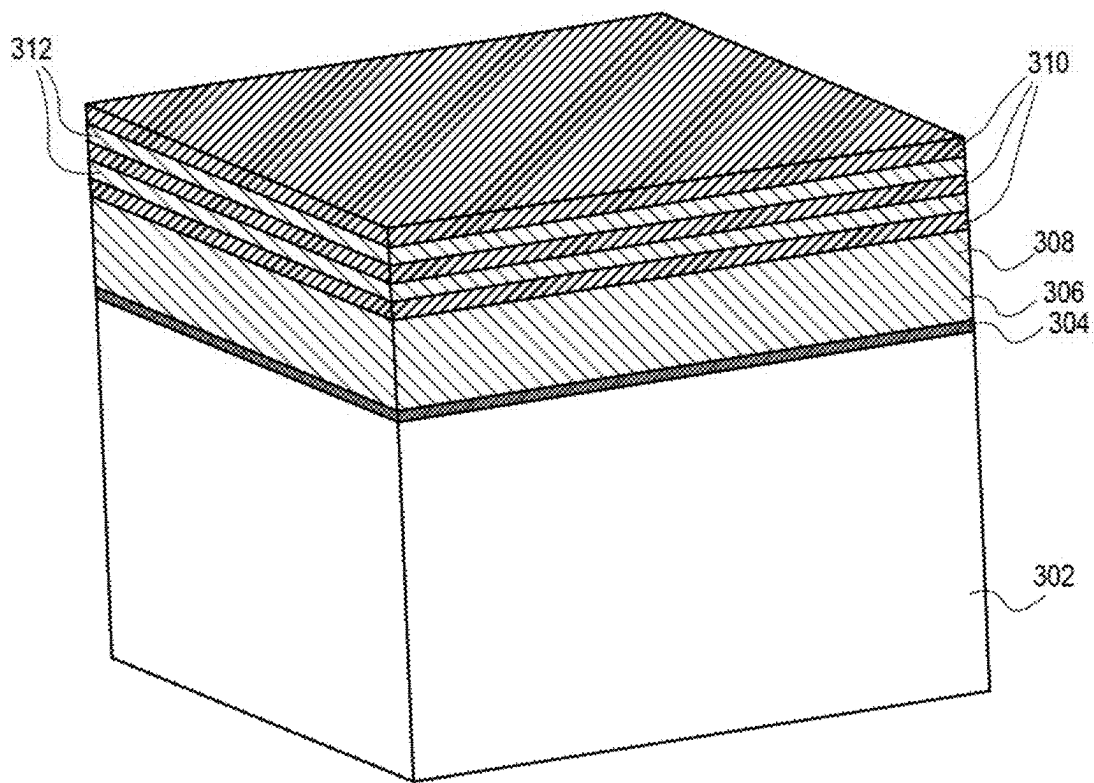
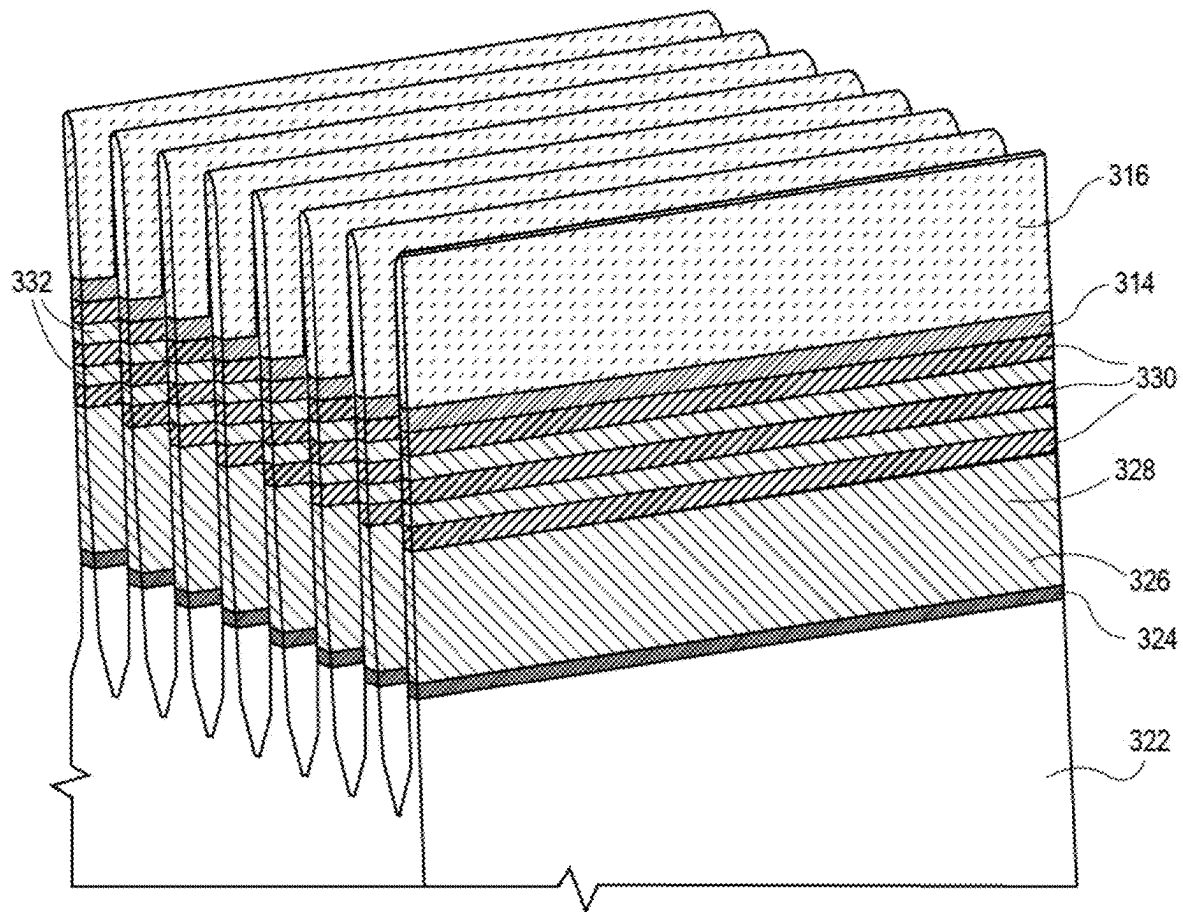


FIG. 3B

**FIG. 3C**

**FIG. 3D**

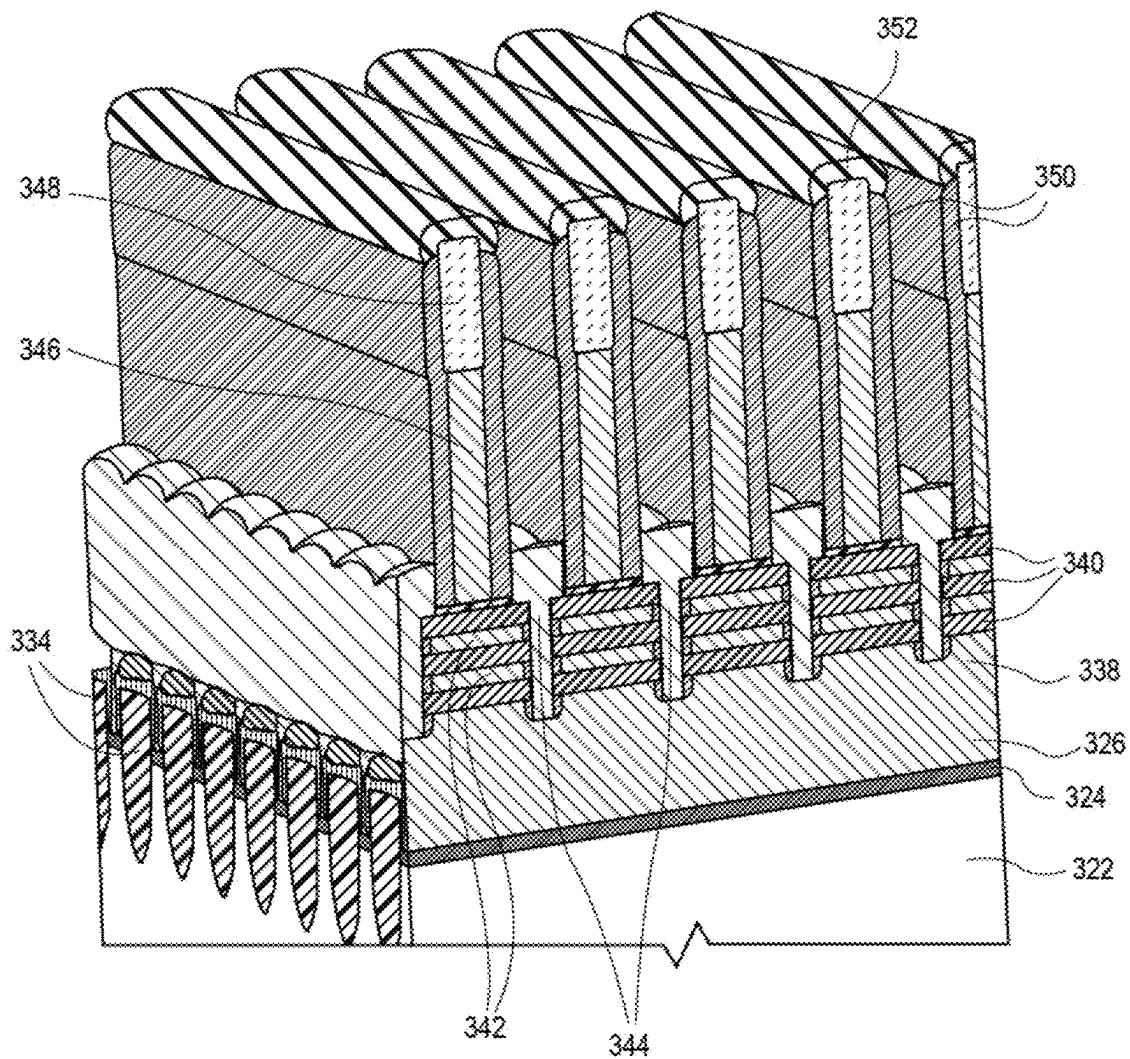
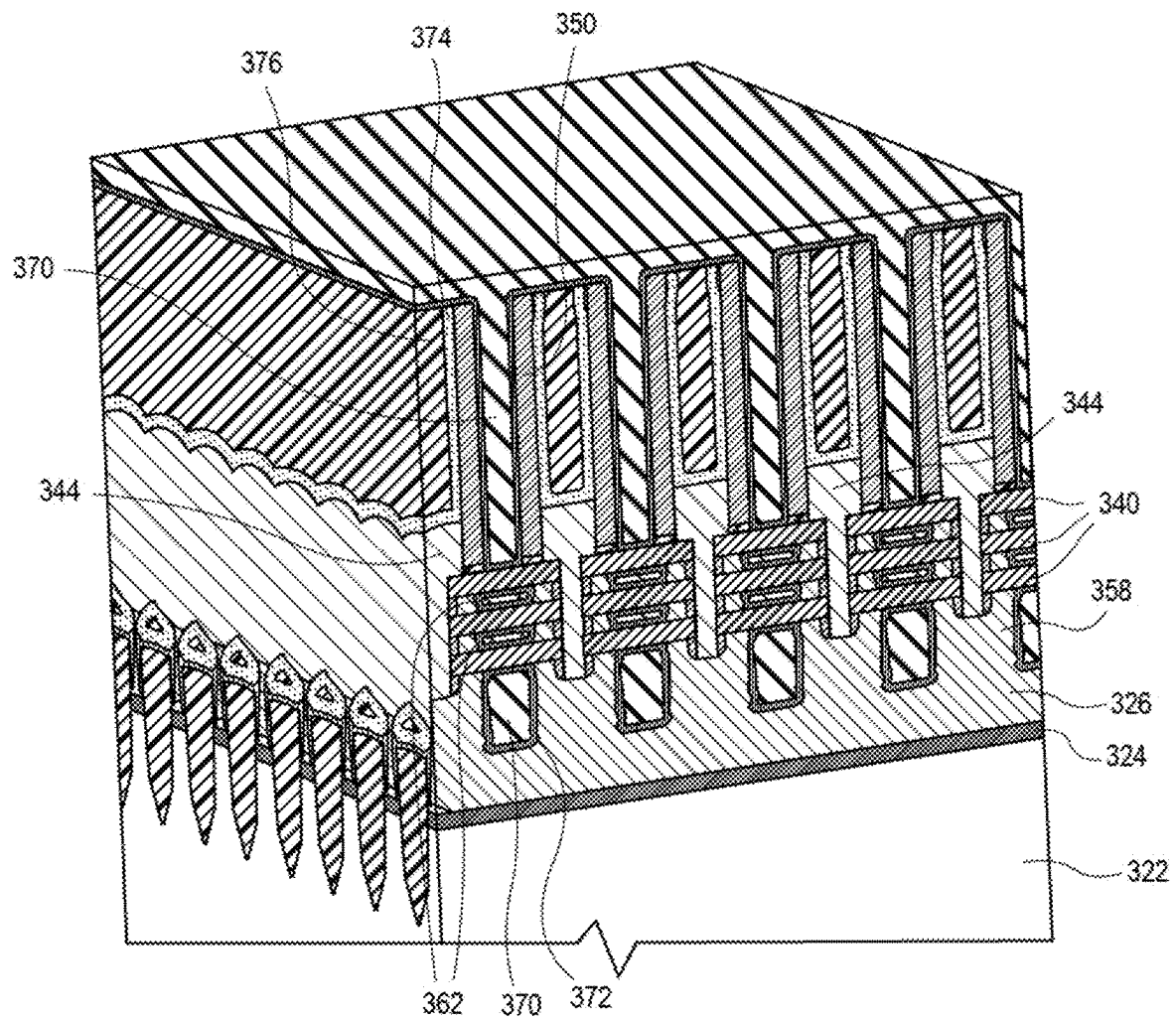


FIG. 3E

**FIG. 3F**

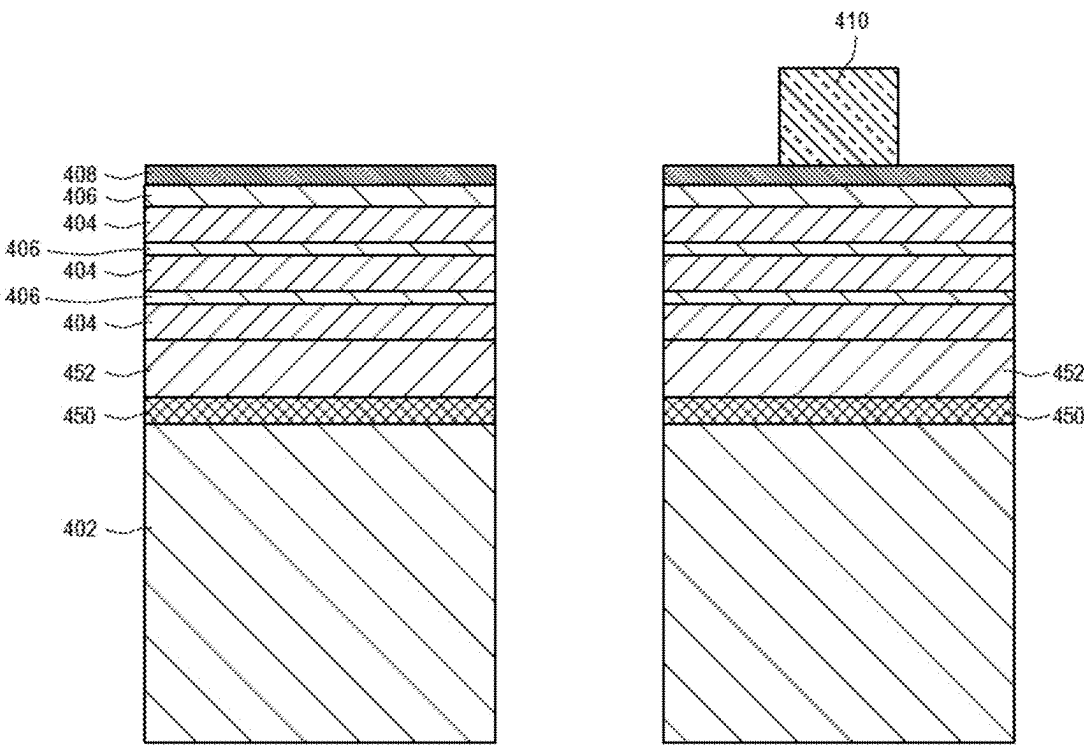


FIG. 4A

FIG. 4B

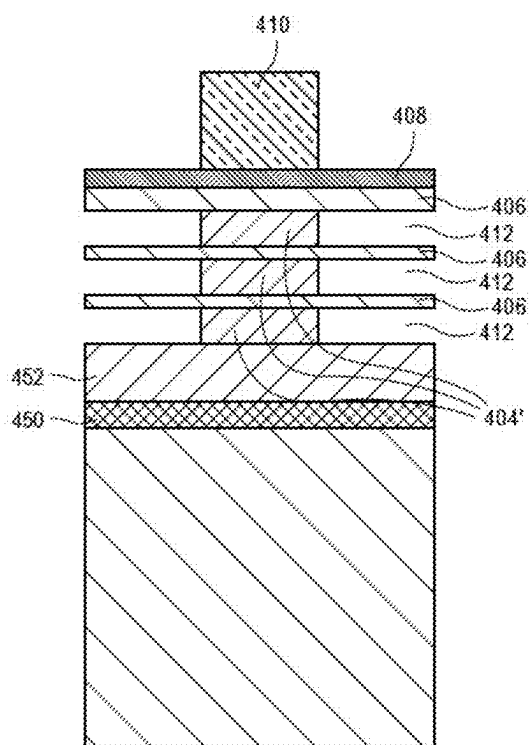


FIG. 4C

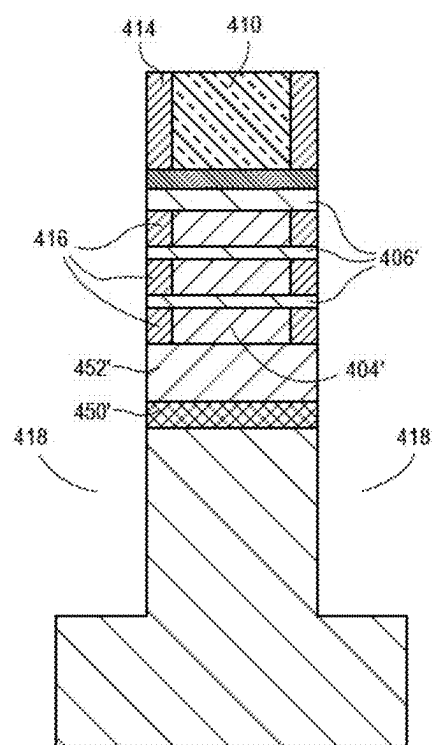


FIG. 4D

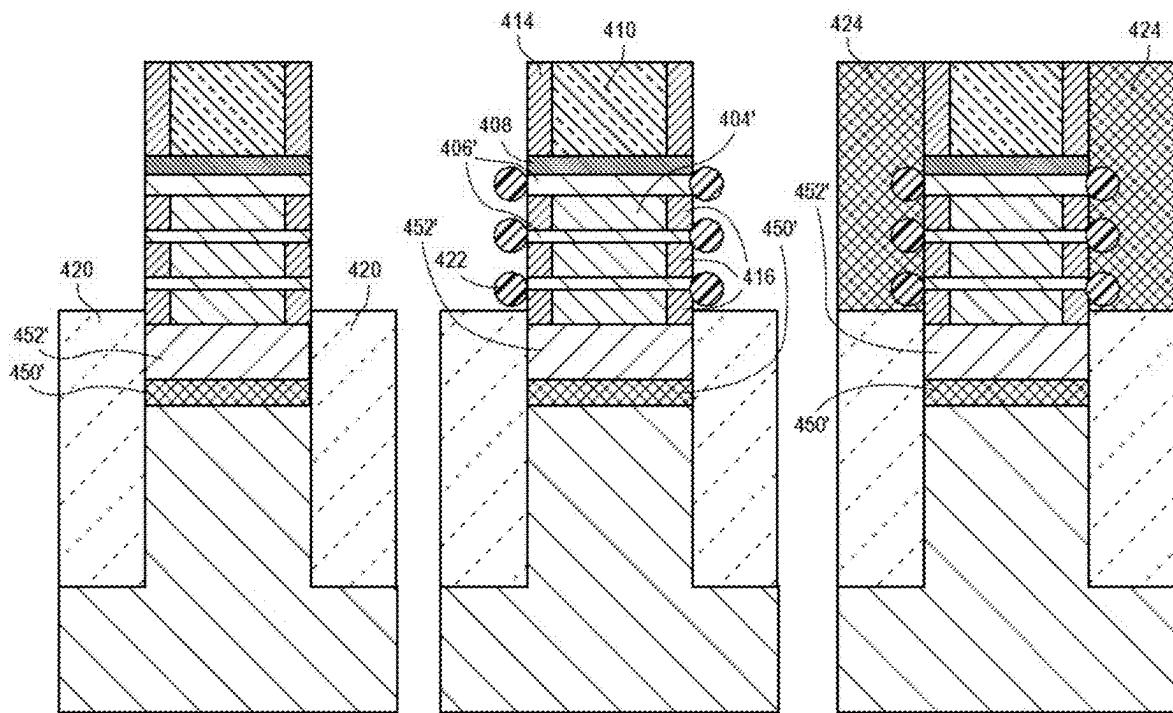


FIG. 4E

FIG. 4F

FIG. 4G

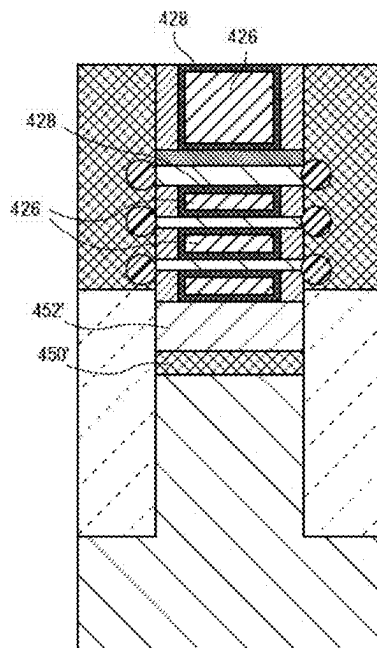


FIG. 4H

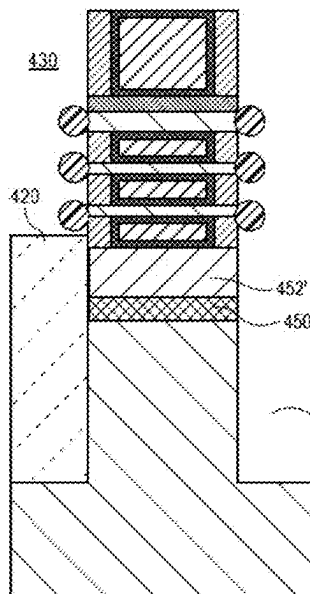


FIG. 4I

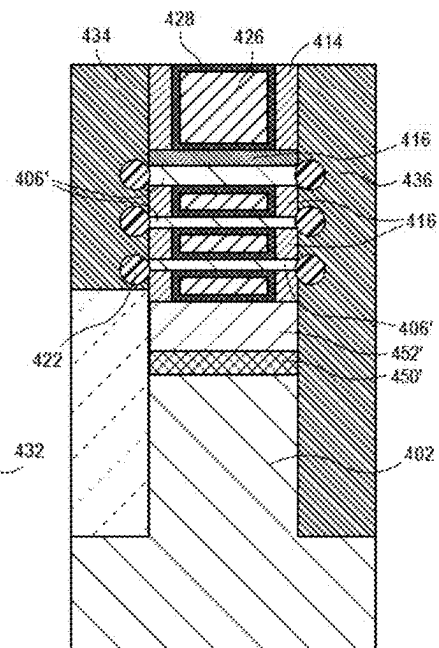


FIG. 4J

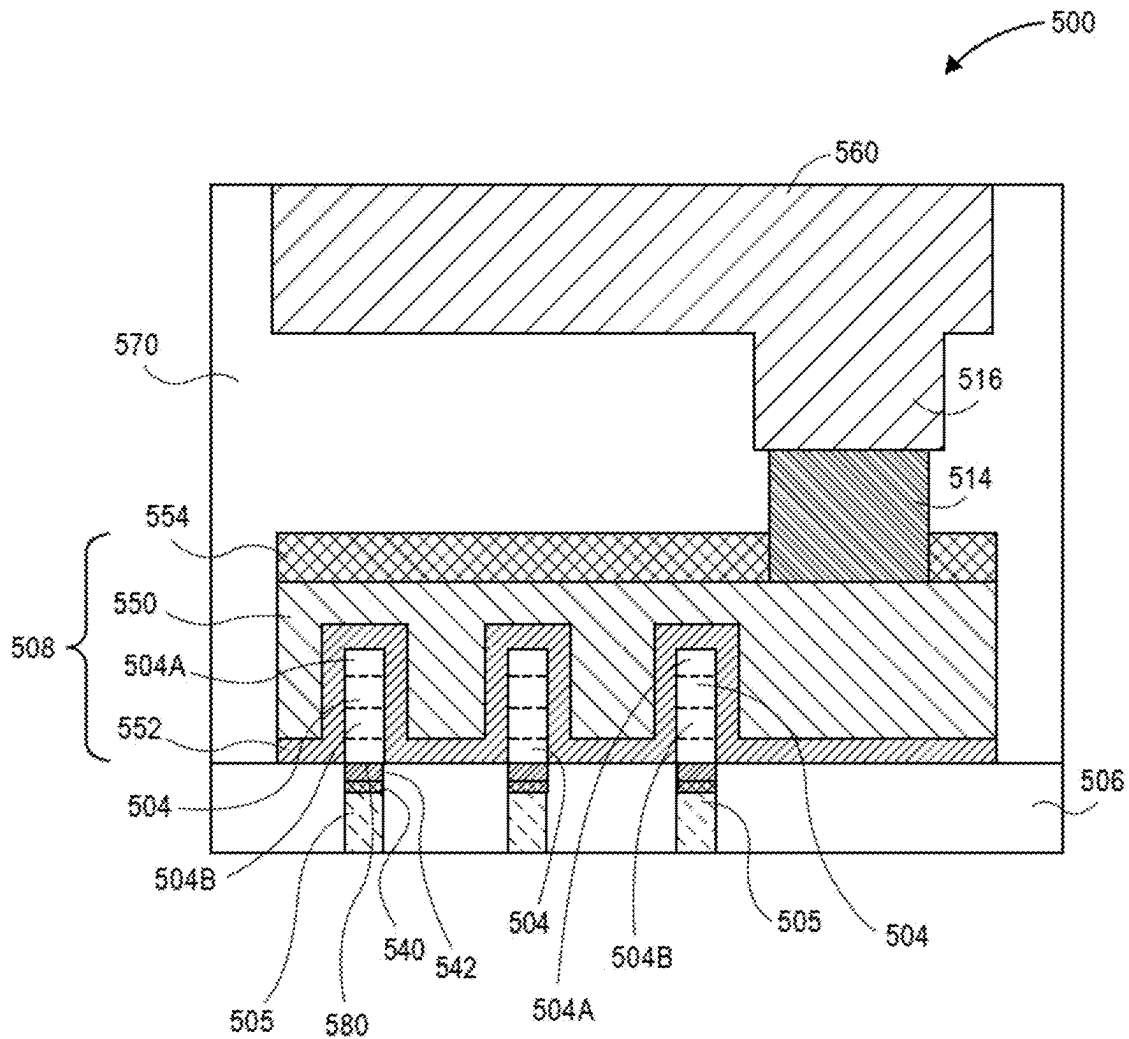


FIG. 5

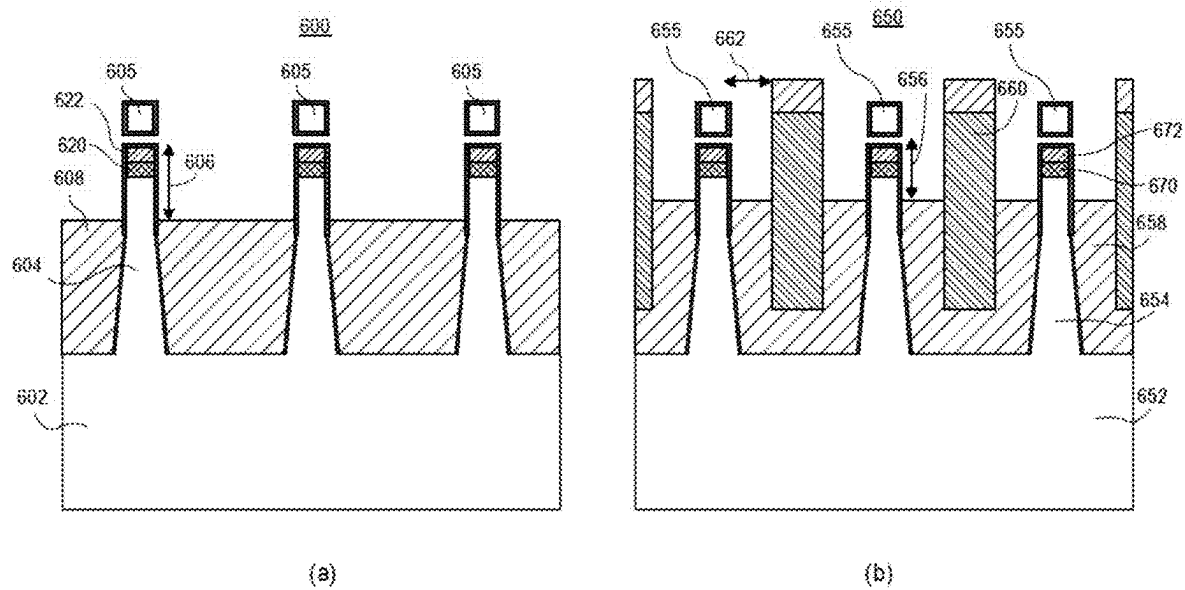


FIG. 6

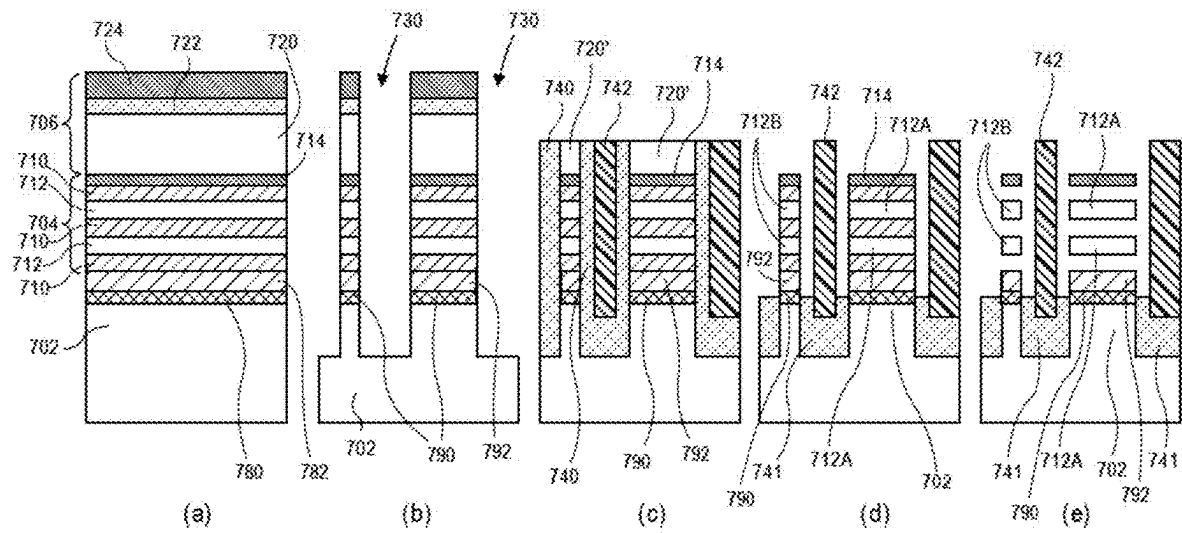


FIG. 7

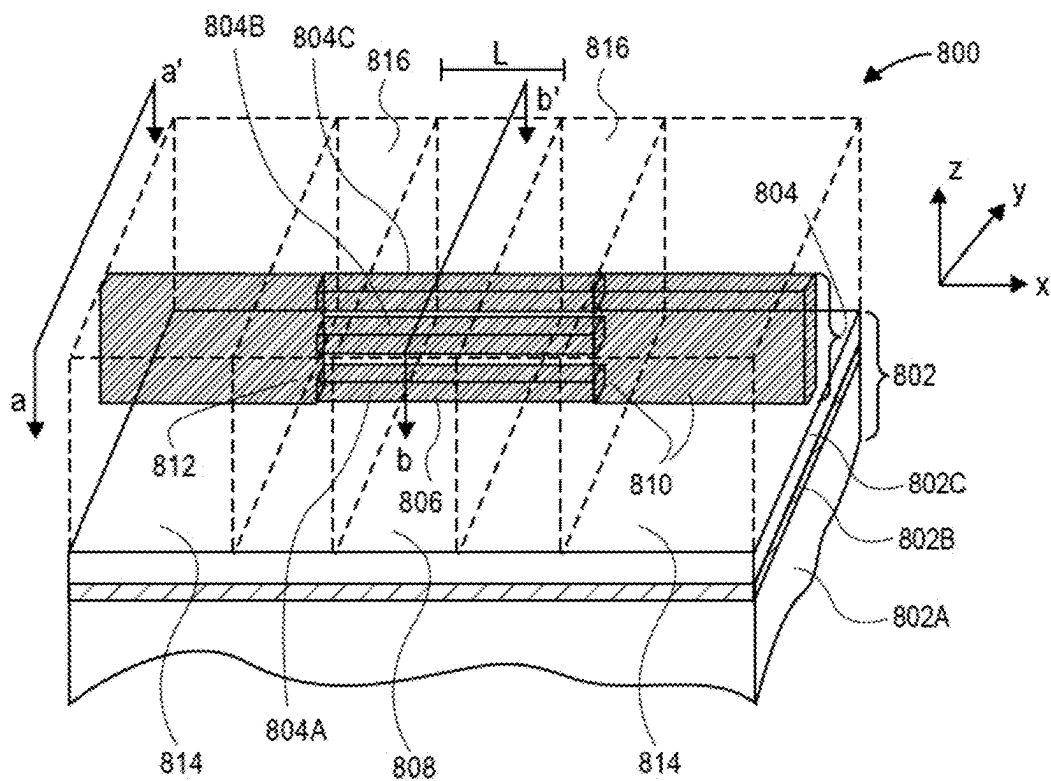


FIG. 8A

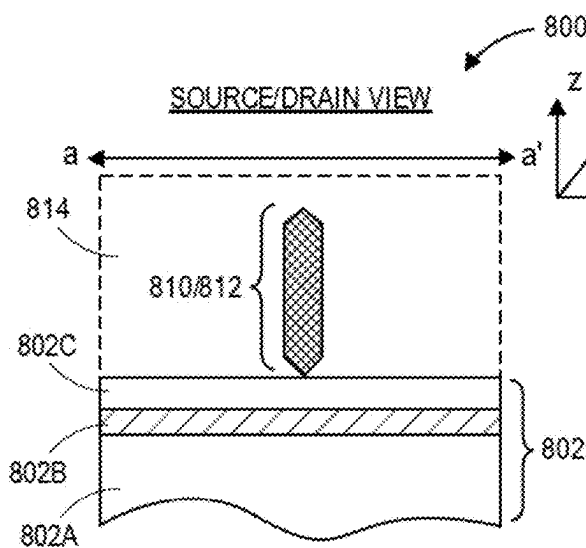


FIG. 8B

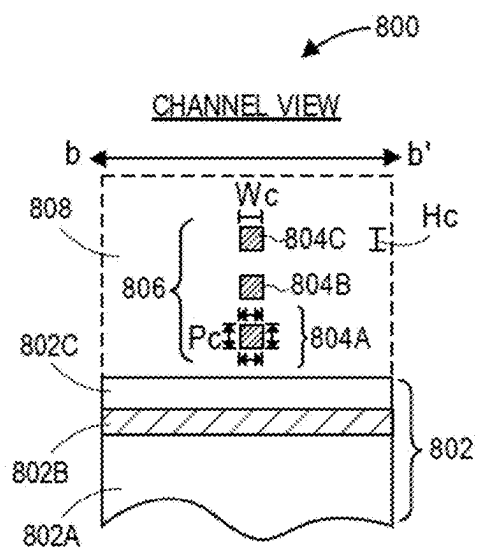
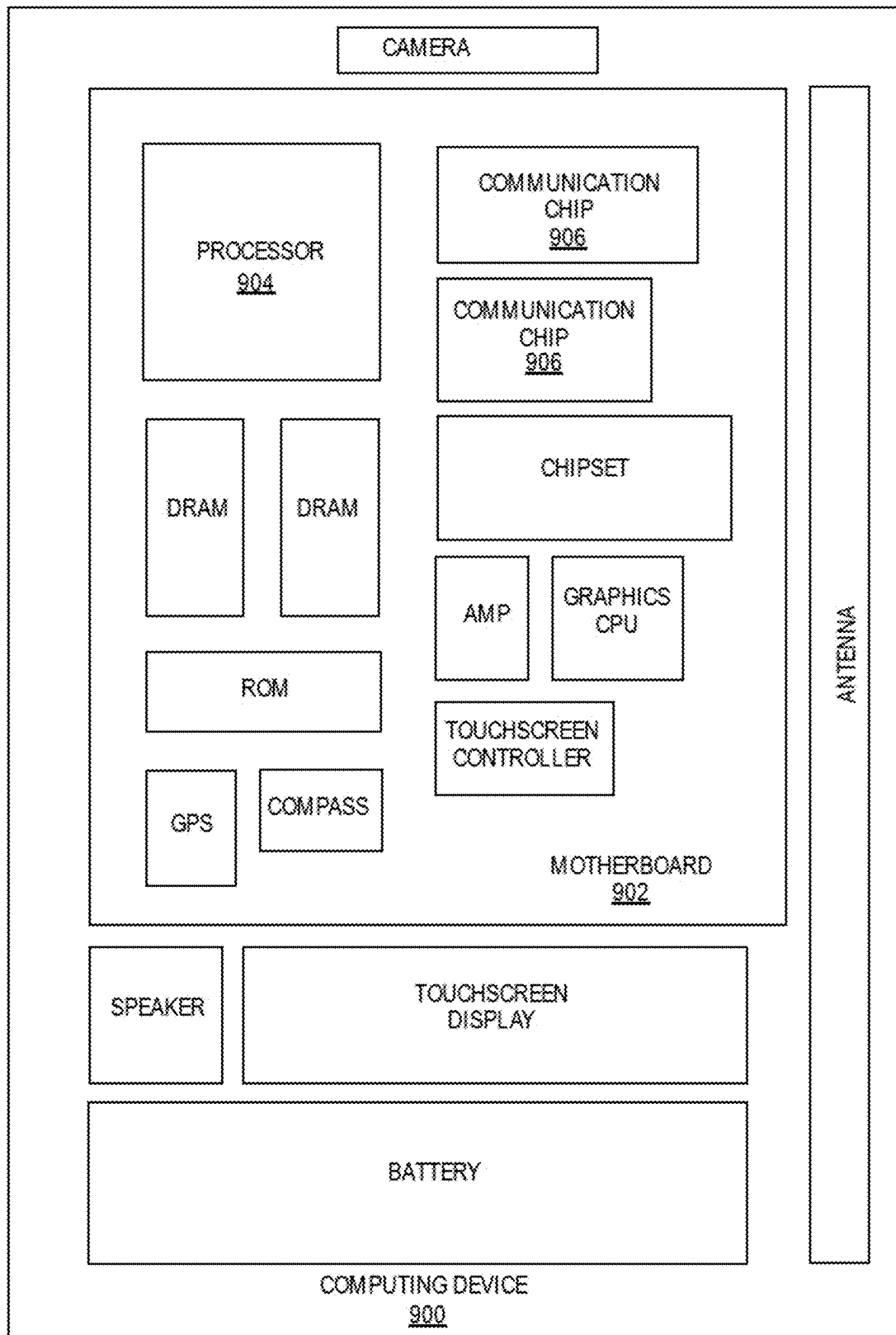


FIG. 8C

**FIG. 9**

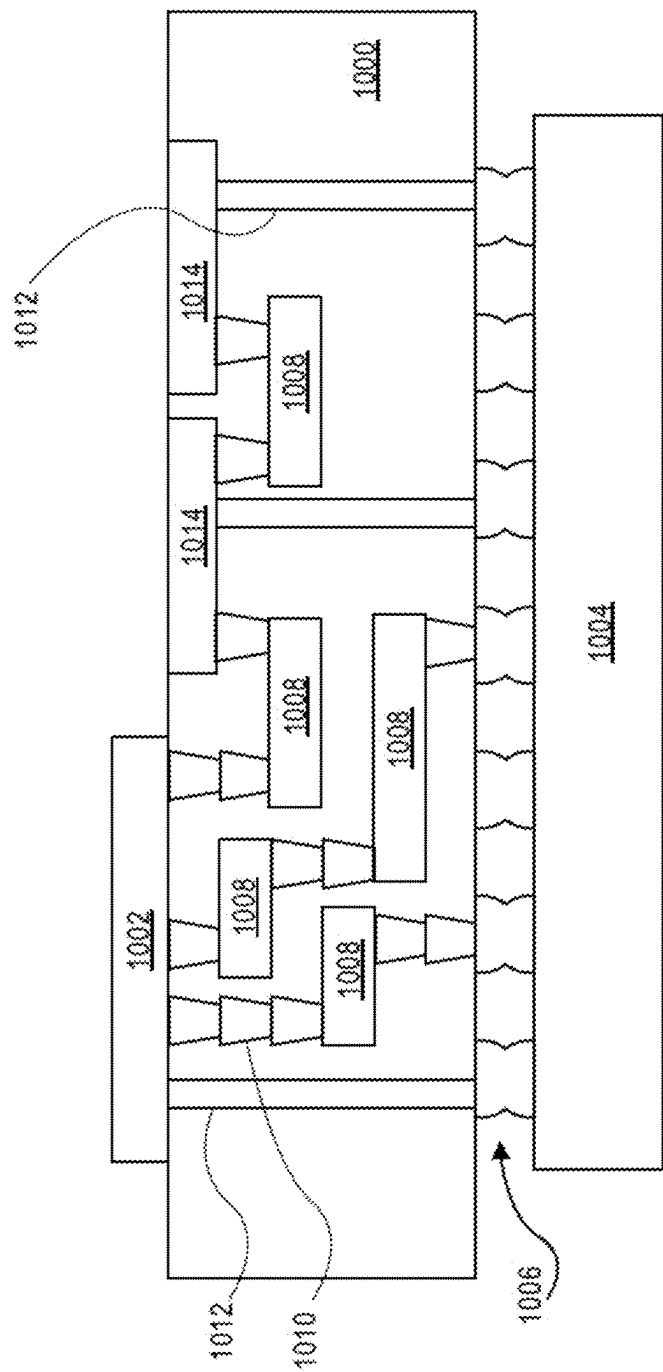


FIG. 10

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GATE-ALL-AROUND INTEGRATED CIRCUIT STRUCTURES HAVING DEVICES WITH CHANNEL-TO-SUBSTRATE ELECTRICAL CONTACT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/719,222, filed on Dec. 18, 2019, the entire contents of which is hereby incorporated by reference herein.

TECHNICAL FIELD

Embodiments of the disclosure are in the field of integrated circuit structures and processing and, in particular, gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact.

BACKGROUND

For the past several decades, the scaling of features in integrated circuits has been a driving force behind an ever-growing semiconductor industry. Scaling to smaller and smaller features enables increased densities of functional units on the limited real estate of semiconductor chips. For example, shrinking transistor size allows for the incorporation of an increased number of memory or logic devices on a chip, lending to the fabrication of products with increased capacity. The drive for ever-more capacity, however, is not without issue. The necessity to optimize the performance of each device becomes increasingly significant.

In the manufacture of integrated circuit devices, multi-gate transistors, such as tri-gate transistors, have become more prevalent as device dimensions continue to scale down. In conventional processes, tri-gate transistors are generally fabricated on either bulk silicon substrates or silicon-on-insulator substrates. In some instances, bulk silicon substrates are preferred due to their lower cost and because they enable a less complicated tri-gate fabrication process. In another aspect, maintaining mobility improvement and short channel control as microelectronic device dimensions scale below the 10 nanometer (nm) node provides a challenge in device fabrication. Nanowires used to fabricate devices provide improved short channel control.

Scaling multi-gate and nanowire transistors has not been without consequence, however. As the dimensions of these fundamental building blocks of microelectronic circuitry are reduced and as the sheer number of fundamental building blocks fabricated in a given region is increased, the constraints on the lithographic processes used to pattern these building blocks have become overwhelming. In particular, there may be a trade-off between the smallest dimension of a feature patterned in a semiconductor stack (the critical dimension) and the spacing between such features.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a cross-sectional view of a gate-all-around device of a gate-all-around integrated circuit structure, in accordance with an embodiment of the present disclosure.

FIG. 2A illustrates a cross-sectional view of a device with channel-to-substrate electrical contact for including in a gate-all-around integrated circuit structure having a device

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with channel-to-substrate electrical contact, in accordance with an embodiment of the present disclosure.

FIG. 2B illustrates a cross-sectional view of another device with channel-to-substrate electrical contact for including in a gate-all-around integrated circuit structure having a device with channel-to-substrate electrical contact, in accordance with another embodiment of the present disclosure.

FIG. 2C illustrates cross-sectional views of exemplary diode structures, in accordance with an embodiment of the present disclosure.

FIGS. 3A-3F illustrate cross-sectional views representing various operations in a method of fabricating a gate-all-around integrated circuit structure, in accordance with an embodiment of the present disclosure.

FIGS. 4A-4J illustrates cross-sectional views of various operations in a method of fabricating a gate-all-around integrated circuit structure, in accordance with an embodiment of the present disclosure.

FIG. 5 illustrates a cross-sectional view of a non-planar integrated circuit structure as taken along a gate line, in accordance with an embodiment of the present disclosure.

FIG. 6 illustrates cross-sectional views taken through nanowires and fins for a non-endcap architecture (left-hand side (a)) versus a self-aligned gate endcap (SAGE) architecture (right-hand side (b)), in accordance with an embodiment of the present disclosure.

FIG. 7 illustrates cross-sectional views representing various operations in a method of fabricating a self-aligned gate endcap (SAGE) structure with gate-all-around devices, in accordance with an embodiment of the present disclosure.

FIG. 8A illustrates a three-dimensional cross-sectional view of a nanowire-based integrated circuit structure, in accordance with an embodiment of the present disclosure.

FIG. 8B illustrates a cross-sectional source or drain view of the nanowire-based integrated circuit structure of FIG. 8A, as taken along the a-a' axis, in accordance with an embodiment of the present disclosure.

FIG. 8C illustrates a cross-sectional channel view of the nanowire-based integrated circuit structure of FIG. 8A, as taken along the b-b' axis, in accordance with an embodiment of the present disclosure.

FIG. 9 illustrates a computing device in accordance with one implementation of an embodiment of the disclosure.

FIG. 10 illustrates an interposer that includes one or more embodiments of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

Gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact are described. In the following description, numerous specific details are set forth, such as specific integration and material regimes, in order to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to one skilled in the art that embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known features, such as integrated circuit design layouts, are not described in detail in order to not unnecessarily obscure embodiments of the present disclosure. Furthermore, it is to be appreciated that the various embodiments shown in the Figures are illustrative representations and are not necessarily drawn to scale.

Certain terminology may also be used in the following description for the purpose of reference only, and thus are not intended to be limiting. For example, terms such as “upper”, “lower”, “above”, and “below” refer to directions

in the drawings to which reference is made. Terms such as “front”, “back”, “rear”, and “side” describe the orientation and/or location of portions of the component within a consistent but arbitrary frame of reference which is made clear by reference to the text and the associated drawings describing the component under discussion. Such terminology may include the words specifically mentioned above, derivatives thereof, and words of similar import.

Embodiments described herein may be directed to front-end-of-line (FEOL) semiconductor processing and structures. FEOL is the first portion of integrated circuit (IC) fabrication where the individual devices (e.g., transistors, capacitors, resistors, etc.) are patterned in the semiconductor substrate or layer. FEOL generally covers everything up to (but not including) the deposition of metal interconnect layers. Following the last FEOL operation, the result is typically a wafer with isolated transistors (e.g., without any wires).

Embodiments described herein may be directed to back-end of line (BEOL) semiconductor processing and structures. BEOL is the second portion of IC fabrication where the individual devices (e.g., transistors, capacitors, resistors, etc.) are interconnected with wiring on the wafer, e.g., the metallization layer or layers. BEOL includes contacts, insulating layers (dielectrics), metal levels, and bonding sites for chip-to-package connections. In the BEOL part of the fabrication stage contacts (pads), interconnect wires, vias and dielectric structures are formed. For modern IC processes, more than 10 metal layers may be added in the BEOL.

Embodiments described below may be applicable to FEOL processing and structures, BEOL processing and structures, or both FEOL and BEOL processing and structures. In particular, although an exemplary processing scheme may be illustrated using a FEOL processing scenario, such approaches may also be applicable to BEOL processing. Likewise, although an exemplary processing scheme may be illustrated using a BEOL processing scenario, such approaches may also be applicable to FEOL processing.

One or more embodiments described herein are directed to patterned wire release approaches enabling substrate contact for nanowire or nanoribbon transistor architectures. One or more embodiments described herein are directed to the fabrication of a 3-terminal bipolar junction transistor (BJT) or a 2-terminal diode for thermal sensing applications.

To provide context, for typical nanowire and nanoribbon device fabrication, the active transistor semiconductor does not have electrical contact to an underlying substrate, such as a silicon fin portion protruding from a bulk silicon substrate. The fabrication of traditional finFET charging protection, such as finFET-based electrostatic discharge device (ESD) diodes, finFET-based BJT or finFET-based thermal diodes, does not translate to nanowire or nanoribbon based technologies. For example, for bulk finFET transistors, the bulk finFET silicon provides a charge dissipating path to facilitate charge and/or thermal discharge. However, there is no known solution for nanowire and/or nanoribbon architectures.

In accordance with one or more embodiments of the present disclosure, addressing issues outlined above, approaches are described for fabricating a connection between NMOS or PMOS nanowires or nanoribbons to a silicon substrate (e.g., fin portion) for dissipating charge during in-process charging (IPC) events and electrostatic discharge (ESD) events. Embodiments may include the retention of a silicon germanium (SiGe) or silicon (Si) release material in a nanostack which would otherwise be

removed during normal device fabrication in order to fabricate a nanowire or nanoribbon structure having channel to substrate electrical contact. The retained SiGe or Si forms the contact to substrate. Embodiments described herein may be implemented to ensure certain transistors are properly protected during IPC and ESD events to ensure circuit functionality.

One or more embodiments described herein include a fabrication process involving providing a pattern-intended region to retain SiGe or Si in the stack of a nanosheet intact, hence contacting the silicon substrate, during nanowire release (SiGe or Si removal). Such embodiments may be as described below in association with FIG. 2A. One or more embodiments described herein include a fabrication process involving providing a pattern-intended region to retain SiGe or Si in the stack of a nanosheet in a dummy gate region intact during dummy gate removal to allow nanowire-to-silicon substrate contact. This latter approach, allows for normal epitaxial growth in source or drain regions to provide for low resistance epitaxial structures having good electrical contact in the source or drain regions. Such embodiments may be as described below in association with FIG. 2B.

As an exemplary device that does not include a channel-to-substrate electrical contact, FIG. 1 illustrates a cross-sectional view of a gate-all-around device of a gate-all-around integrated circuit structure, in accordance with an embodiment of the present disclosure.

Referring to FIG. 1, an integrated circuit structure **100** includes a semiconductor substrate **102** having a fin **104** protruding therefrom. The substrate **102** may be a bulk silicon substrate, and the fin **104** may be a sub-fin structure of a nanowire or nanoribbon device. Nanowires or nanoribbons **106** are above the fin **104** and may be stacked as a vertical arrangement of horizontal nanowires or nanoribbons, as is depicted. A gate stack **108** surrounds channel regions of the nanowires or nanoribbons **106**. Source or drain structures **110** are on either side of the gate stack **108**. Source or drain contacts **112** are on the source or drain structures **110**. Region **114** collectively represents isolation or dielectric layers or features integrated around and between the above described device features. The integrated circuit structure **100** does not include a channel to substrate electrical contact. Accordingly, the integrated circuit structure **100** may represent an example of a standard nanowire or nanoribbon device.

As a first exemplary device that includes a channel-to-substrate electrical contact, FIG. 2A illustrates a cross-sectional view of a device with channel-to-substrate electrical contact for including in a gate-all-around integrated circuit structure having a device with channel-to-substrate electrical contact, in accordance with an embodiment of the present disclosure.

Referring to FIG. 2A, an integrated circuit structure **200** includes a semiconductor substrate **202** having a fin **204** protruding therefrom. The substrate **202** may be a bulk silicon substrate, and the fin **204** may be a sub-fin structure of a nanowire or nanoribbon device. Nanowires or nanoribbons **206** are above the fin **204** and may be stacked as a vertical arrangement of horizontal nanowires or nanoribbons, as is depicted. Intervening semiconductor layers **205** are between and beneath the nanowires or nanoribbons **206**. A gate stack **208** surrounds channel regions of the nanowires or nanoribbons **206** as well as portions of the semiconductor layers **205** between and beneath the nanowires or nanoribbons **206**. Source or drain structures **210** are on either side of the gate stack **208**. Source or drain contacts **212** are on the source or drain structures **210**. Region **214** collectively

represents isolation or dielectric layers or features integrated around and between the above described device features. The integrated circuit structure **200** includes a channel-to-substrate electrical contact via portions of the semiconductor layers **205** between and beneath the nanowires or nanoribbons **206**. The integrated circuit structure **200** may be fabricated by inhibiting nanowire release for certain devices during fabrication of standard nanowire or nanoribbon devices.

In an embodiment, an integrated circuit structure includes a device of the type **200** of FIG. 2A together with a device of the type **100** of FIG. 1, e.g., in a same circuit. In an exemplary embodiment, an integrated circuit structure includes a first vertical arrangement of horizontal nanowires **206** above a first fin **204**. A channel region of the first vertical arrangement of horizontal nanowires **206** is electrically coupled to the first fin **204** by a semiconductor material layer (**205**) directly between the first vertical arrangement of horizontal **206** nanowires and the first fin **204**. A first gate stack **208** is over the first vertical arrangement of horizontal nanowires **206**. A second vertical arrangement of horizontal nanowires **106** is above a second fin **104**. A channel region of the second vertical arrangement of horizontal nanowires **106** is electrically isolated from the second fin **104**. A second gate stack **108** is over the second vertical arrangement of horizontal nanowires **106**. In one embodiment, the semiconductor material layer **205** is between the channel region of the first vertical arrangement of horizontal nanowires **206** and the first fin **204**, and between source or drain regions **210** of the first vertical arrangement of horizontal nanowires **206** and the first fin **204**, as is depicted.

In an embodiment, the integrated circuit structure further includes a second pair of epitaxial source or drain structures **110** at first and second ends of the second vertical arrangement of horizontal nanowires **106**, as is depicted. In one embodiment, the second pair of epitaxial source or drain structures **110** is a pair of non-discrete epitaxial source or drain structures, as is depicted. In another embodiment, the second pair of epitaxial source or drain structures **110** is a pair of discrete epitaxial source or drain structures. In an embodiment, there are no epitaxial source or drain structures at first and second ends of the first vertical arrangement of horizontal nanowires **206**, as is depicted.

In an embodiment, the first **204** and second **104** fins each include a portion of a bulk silicon substrate **202** or **102**, respectively. In an embodiment, the nanowires **206** and **106** of the first and second vertical arrangements of horizontal nanowires include silicon, and the semiconductor material layer **205** includes silicon and germanium. In an embodiment, each of the first **208** and second **108** gate stacks include a high-k gate dielectric layer and a metal gate electrode.

As a second exemplary device that includes a channel-to-substrate electrical contact, FIG. 2B illustrates a cross-sectional view of another device with channel-to-substrate electrical contact for including in a gate-all-around integrated circuit structure having a device with channel-to-substrate electrical contact, in accordance with another embodiment of the present disclosure.

Referring to FIG. 2B, an integrated circuit structure **250** includes a semiconductor substrate **252** having a fin **254** protruding therefrom. The substrate **252** may be a bulk silicon substrate, and the fin **254** may be a sub-fin structure of a nanowire or nanoribbon device. Nanowires or nanoribbons **256** are above the fin **254** and may be stacked as a vertical arrangement of horizontal nanowires or nanoribbons, as is depicted. Intervening semiconductor layers **255**

are between and beneath the nanowires or nanoribbons **256** in a channel location of the nanowires or nanoribbons **256**. A gate stack **258** surrounds channel regions of the nanowires or nanoribbons **256** as well as the semiconductor layers **255** between and beneath the nanowires or nanoribbons **256**. Source or drain structures **260** are on either side of the gate stack **258**. Source or drain contacts **262** are on the source or drain structures **260**. Region **264** collectively represents isolation or dielectric layers or features integrated around and between the above described device features. The integrated circuit structure **250** includes a channel-to-substrate electrical contact via the semiconductor layers **255** between and beneath the nanowires or nanoribbons **256**. The integrated circuit structure **250** may be fabricated by inhibiting nanowire release in a dummy gate replacement process for certain devices during fabrication of standard nanowire or nanoribbon devices. For structure **250**, electrical contact is only made in a region covered by the gate stack **258** (e.g., in a location of an initial dummy gate). This embodiment enables epitaxial source or drain structure formation for low resistance source or drain contact.

In an embodiment, an integrated circuit structure includes a device of the type **250** of FIG. 2B together with a device of the type **100** of FIG. 1, e.g., in a same circuit. In an exemplary embodiment, an integrated circuit structure includes a first vertical arrangement of horizontal nanowires **256** above a first fin **254**. A channel region of the first vertical arrangement of horizontal nanowires **256** is electrically coupled to the first fin **254** by a semiconductor material layer **255** directly between the channel region of the first vertical arrangement of horizontal nanowires **256** and the first fin **254** but not between source or drain regions or locations **260** of the first vertical arrangement of horizontal nanowires **256** and the first fin **254**. A first gate stack **258** is over the first vertical arrangement of horizontal nanowires **256**. A second vertical arrangement of horizontal nanowires **106** is above a second fin **104**. A channel region of the second vertical arrangement of horizontal nanowires **106** is electrically isolated from the second fin **104**. A second gate stack **108** is over the second vertical arrangement of horizontal nanowires **106**.

In an embodiment, the integrated circuit structure further includes a first pair of epitaxial source or drain structures **260** at first and second ends of the first vertical arrangement of horizontal nanowires **256**, as is depicted, and a second pair of epitaxial source or drain structures **110** at first and second ends of the second vertical arrangement of horizontal nanowires **106**, as is depicted. In one such embodiment, the first **260** and second **110** pairs of epitaxial source or drain structures are first and second pairs of non-discrete epitaxial source or drain structures, as is depicted. In another embodiment, not depicted, the first **260** and second **110** pairs of epitaxial source or drain structures are first and second pairs of discrete epitaxial source or drain structures.

In an embodiment, the first **254** and second **104** fins each include a portion of a bulk silicon substrate **252** or **102**, respectively. In an embodiment, the nanowires **256** and **106** of the first and second vertical arrangements of horizontal nanowires include silicon, and the semiconductor material layer **255** includes silicon and germanium. In an embodiment, each of the first **258** and second **108** gate stacks include a high-k gate dielectric layer and a metal gate electrode.

It is to be appreciated that the embodiments described herein can also include other implementations such as nanowires and nanoribbons with various widths, thicknesses and/or materials including but not limited to Si and SiGe.

Embodiments may be applicable for fabricating devices for circuits requiring charge dissipating to a substrate.

In another aspect, it is to be appreciated that large ESD currents can result in extreme temperatures within the diodes ultimately leading to failure. In traditional bulk diodes, shallow trench isolation (STI) structures form a separation between an anode and a cathode, causing a current path to travel into the substrate where the heat generated has a large volume in which to dissipate. In nanoribbon diodes (e.g., structure **270** of FIG. 2C, described below), current flow, and thus heat generation, occurs in the nanoribbons which is a much smaller volume and is thus susceptible to early failure of the device. In an embodiment, a path of current flow to a substrate is provided by retaining SiGe in a gate region between the Si wires, enabling larger diode junction area and enhanced heat dissipation.

To provide further context, a traditional STI based BJT uses an extra P+ emitter implant to tune emitter-base junction characteristics which in turn modulate the BJT performance metrics. Nanoribbon (or nanowire) structures do not facilitate fabrication of a 3-terminal efficient BJT. However, the application of embodiments described herein can enable the fabrication of structures that can achieve performance comparable to STI based BJTs in nanoribbon (or nanowire) technology. It is to be appreciated that a BJT may be fabricated according to embodiments described herein to provide performance sufficient to function as a thermal sensor.

FIG. 2C illustrates cross-sectional views of exemplary diode structures, in accordance with an embodiment of the present disclosure.

Referring to FIG. 2C, a nanoribbon diode **270** includes an N-Well **272**. Nanoribbon stacks **274** are on the N-Well **272**. N-type source or drain structures **276** and P-type source or drain structures **278** are between the nanoribbon stacks **274**.

Referring again to FIG. 2C, a well diode **280** includes an N-Well **282**. A cathode **283A** and an anode **283B** are above the N-Well **282**. The cathode **283A** includes nanoribbon or nanowire stacks **286A** including intervening semiconductor material to provide channel to N-Well **282** contact. Gate stacks **288A** are over the nanoribbon or nanowire stacks **286A**. N-type source or drain structures **284A** are between the gate stacks **288A**. The anode **283B** includes nanoribbon or nanowire stacks **286B** including intervening semiconductor material to provide channel to N-Well **282** contact. Gate stacks **288B** are over the nanoribbon or nanowire stacks **286B**. P-type source or drain structures **284B** are between the gate stacks **288B**.

Referring again to FIG. 2C, a well diode **290A** with junction formed inside the well includes an N-Well **291** having a P+ doping region **292** formed therein. A cathode **293A** is above the N-Well **291**, and an anode **293B** is above the P+ doping region **292**. The cathode **293A** includes nanoribbon or nanowire stacks including intervening semiconductor material to provide channel to N-Well **291** contact. Gate stacks are over the nanoribbon or nanowire stacks. N-type source or drain structures **294A** are between the gate stacks. The anode **293B** includes nanoribbon or nanowire stacks including intervening semiconductor material to provide channel to P+ doping region **292** contact. Gate stacks are over the nanoribbon or nanowire stacks. P-type source or drain structures **294B** are between the gate stacks.

Referring again to FIG. 2C, a 3-terminal band gap diode **290B** includes a P-Well **296**, and an N-Well **295** having a P+ doping region **297** formed therein. A collector **298A** is above the P-Well **296**, a base **298B** is above the N-Well **295**, and an emitter **298C** is above the P+ doping region **297**. The

collector **298A** includes nanoribbon or nanowire stacks including intervening semiconductor material to provide channel to P-Well **296** contact. Gate stacks are over the nanoribbon or nanowire stacks. P-type source or drain structures **299B** are between the gate stacks. The base **298B** includes nanoribbon or nanowire stacks including intervening semiconductor material to provide channel to N-Well **295** contact. Gate stacks are over the nanoribbon or nanowire stacks. N-type source or drain structures **299A** are between the gate stacks. The emitter **298C** includes nanoribbon or nanowire stacks including intervening semiconductor material to provide channel to P+ doping region **297**. Gate stacks are over the nanoribbon or nanowire stacks. P-type source or drain structures **298B** are between the gate stacks.

It is to be appreciated, in a particular embodiment, channel layers may be silicon, and intervening layers may be silicon germanium. As used throughout, a silicon layer may be used to describe a silicon material composed of a very substantial amount of, if not all, silicon. However, it is to be appreciated that, practically, 100% pure Si may be difficult to form and, hence, could include a tiny percentage of carbon, germanium or tin. Such impurities may be included as an unavoidable impurity or component during deposition of Si or may "contaminate" the Si upon diffusion during post deposition processing. As such, embodiments described herein directed to a silicon layer may include a silicon layer that contains a relatively small amount, e.g., "impurity" level, non-Si atoms or species, such as Ge, C or Sn. It is to be appreciated that a silicon layer as described herein may be undoped or may be doped with dopant atoms such as boron, phosphorous or arsenic.

As used throughout, a silicon germanium layer may be used to describe a silicon germanium material composed of substantial portions of both silicon and germanium, such as at least 5% of both. In some embodiments, the amount of germanium is greater than the amount of silicon. In particular embodiments, a silicon germanium layer includes approximately 60% germanium and approximately 40% silicon (Si₄₀Ge₆₀). In other embodiments, the amount of silicon is greater than the amount of germanium. In particular embodiments, a silicon germanium layer includes approximately 30% germanium and approximately 70% silicon (Si₇₀Ge₃₀). It is to be appreciated that, practically, 100% pure silicon germanium (referred to generally as SiGe) may be difficult to form and, hence, could include a tiny percentage of carbon or tin. Such impurities may be included as an unavoidable impurity or component during deposition of SiGe or may "contaminate" the SiGe upon diffusion during post deposition processing. As such, embodiments described herein directed to a silicon germanium layer may include a silicon germanium layer that contains a relatively small amount, e.g., "impurity" level, non-Ge and non-Si atoms or species, such as carbon or tin. It is to be appreciated that a silicon germanium layer as described herein may be undoped or may be doped with dopant atoms such as boron, phosphorous or arsenic.

Described below are various devices and processing schemes that may be used to fabricate a device without a channel to substrate contact that can be integrated with a device having a channel to substrate contact. It is to be appreciated that the exemplary embodiments need not necessarily require all features described, or may include more features than are described.

As an exemplary process flow for fabricating a gate-all-around device of a gate-all-around integrated circuit structure having a device with channel-to-substrate electrical

contact, FIGS. 3A-3F illustrate cross-sectional views representing various operations in a method of fabricating a gate-all-around integrated circuit structure, in accordance with an embodiment of the present disclosure.

Referring to FIG. 3A, a starting structure may include a defect modification layer **304** formed on a substrate **302**. In one embodiment, during processing, a silicon (Si) substrate **302** is first modified to allow subsequent buffer layers to easily relax, as well as to trap defects that otherwise promote relaxation below what will become the channel material. In one embodiment, the defect modification layer **304** is a layer including ion implant damage or is a defect-rich Si growth layer, or a combination thereof. In another embodiment, the layer **304** is a defect-rich SiGe layer.

Referring to FIG. 3B, a relaxed buffer layer **306** is grown on the defect modification layer **304**. In an embodiment, the relaxed buffer layer **306** is a relaxed Si₇₀Ge₃₀ layer.

Referring to FIG. 3C, a sacrificial layer **308** is grown on the relaxed buffer layer **306**. In an embodiment, the sacrificial layer **308** is a Si₇₀Ge₃₀ layer. In one embodiment, the sacrificial layer **308** has a composition the same as or substantially the same as the relaxed buffer layer **306**. In a particular embodiment, the sacrificial layer **308** is a Si₇₀Ge₃₀ layer and the relaxed buffer layer **306** is a relaxed Si₇₀Ge₃₀ layer. In one embodiment, the sacrificial layer **308** is formed by continuing growth relaxed buffer layer **306** in a same deposition process, and the sacrificial layer **308** and growth relaxed buffer layer **306** may appear as a unified layer. In another embodiment, the sacrificial layer **308** has a composition different from the relaxed buffer layer **306**.

Alternating channel layers **310** and intervening sacrificial layers **312** are formed on the sacrificial layer **308**. In an embodiment, the channel layers **310** are Si₄₀Ge₆₀ channel layers. In an embodiment, the intervening sacrificial layers **312** are intervening Si₇₀Ge₃₀ layer.

Referring to FIG. 3D, the stack of materials of FIG. 3D have a patterned mask **314/316** formed thereon, which may include a hardmask portion **316** and an etch stop portion **314**. The patterned mask **314/316** is used to etch a plurality of fins in the stack of materials of FIG. 3D. In an embodiment, the etching is deeper than the defect modification layer **304**. In one such embodiment, each fin includes a patterned substrate portion **322**, a defect modification layer **324**, a buffer layer **326**, a sacrificial layer **328**, channel layers **330**, and intervening sacrificial layers **332**.

Referring to FIG. 3E, isolation structures **334**, such as shallow trench isolation structures, are formed between fins. Dummy gate structures are then formed over the fins and over the isolation structures **334**. Each of the dummy gate structures includes a dummy gate electrode **346**, a hardmask **348**, sidewall spacers **350**, and a dielectric helmet **352**. A dummy gate dielectric may also be included beneath the dummy gate electrode **346**, as is depicted. In one embodiment, the dummy gate dielectric is a remnant of the mask layer **314**.

Referring again to FIG. 3E, the dummy gate structures are used as a mask to etch trenches into exposed portions of the fins, e.g., into source and drain regions of the fins. The etching removes portions of channel layers **330** to form channel layers **340**, and removes portions of the intervening sacrificial layers **332** to form intervening sacrificial layers **342**. In one embodiment, the etching extends at least partially into sacrificial layer **328** to form recessed sacrificial layer **338**.

Referring again to FIG. 3E, epitaxial source or drain structures **344** are then grown in the trenches. In an embodiment, the epitaxial source or drain structures provide strain for the channel layers **340**.

Referring to FIG. 3F, following source/drain deposition, the dummy gate electrode **346** and the sacrificial layers **342** are removed and replaced with a permanent gate electrode **370** and gate dielectric stack **372**. In one embodiment, the gate electrode **370** is a metal gate electrode, and the gate dielectric **372** is a high-k gate dielectric. In an embodiment, a portion of the recessed sacrificial layer **338** is further recessed in the gate channel region to form a gate electrode **370** in a further recessed sacrificial layer **358**, where the gate electrode **370** is deeper than the epitaxial source or drain structures **344**, as is depicted. In an embodiment, portions **362** of the sacrificial layers **342** are retained on either side of the gate electrode **370**, as is also depicted. In an embodiment, a contact barrier layer **374** and conductive fill **376** are formed over the epitaxial source or drain structures **344**. It is to be appreciated that the structure of FIG. 3F may be further planarized in order to confine the permanent gate material **370** to the gate locations.

In an embodiment, the structure of FIG. 3F is included in an integrated circuit structure that also includes a device with channel-to-substrate electrical contact such as described in association with FIG. 2A or FIG. 2B. The device with channel-to-substrate electrical contact may be fabricated together with the structure of FIG. 3F by including process operations that preserve a portion of the sacrificial layer **328** and the intervening sacrificial layers **332** at least in a location beneath a gate electrode.

In another aspect, nanowire release processing may be performed through a replacement gate trench. Examples of such release processes are described below. Additionally, in yet another aspect, backend (BE) interconnect scaling can result in lower performance and higher manufacturing cost due to patterning complexity. Embodiments described herein may be implemented to enable front-side and back-side interconnect integration for nanowire transistors. Embodiments described herein may provide an approach to achieve a relatively wider interconnect pitch. The result may be improved product performance and lower patterning costs. Embodiments may be implemented to enable robust functionality of scaled nanowire or nanoribbon transistors with low power and high performance.

One or more embodiments described herein are directed dual epitaxial (EPI) connections for nanowire or nanoribbon transistors using partial source or drain (SD) and asymmetric trench contact (TCN) depth. In an embodiment, an integrated circuit structure is fabricated by forming source-drain openings of nanowire/nanoribbon transistors which are partially filled with SD epitaxy. A remainder of the opening is filled with a conductive material. Deep trench formation on one of the source or drain side enables direct contact to a back-side interconnect level.

As an exemplary process flow for fabricating a gate-all-around device of a gate-all-around integrated circuit structure having a device with channel-to-substrate electrical contact, FIGS. 4A-4J illustrates cross-sectional views of various operations in a method of fabricating a gate-all-around integrated circuit structure, in accordance with an embodiment of the present disclosure.

Referring to FIG. 4A, a method of fabricating an integrated circuit structure includes forming a starting stack which includes alternating sacrificial layers **404** and nanowires **406** above a fin **402**, such as a silicon fin. The nanowires **406** may be referred to as a vertical arrangement of nanow-

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ires. A protective cap **408** may be formed above the alternating sacrificial layers **404** and nanowires **406**, as is depicted. A relaxed buffer layer **452** and a defect modification layer **450** may be formed beneath the alternating sacrificial layers **404** and nanowires **406**, as is also depicted.

Referring to FIG. **4B**, a gate stack **410** is formed over the vertical arrangement of horizontal nanowires **406**. Portions of the vertical arrangement of horizontal nanowires **406** are then released by removing portions of the sacrificial layers **404** to provide recessed sacrificial layers **404'** and cavities **412**, as is depicted in FIG. **4C**.

It is to be appreciated that the structure of FIG. **4C** may be fabricated to completion without first performing the deep etch and asymmetric contact processing described below. In either case (e.g., with or without asymmetric contact processing), in an embodiment, a fabrication process involves use of a process scheme that provides a gate-all-around integrated circuit structure having epitaxial nubs, which may be vertically discrete source or drain structures.

Referring to FIG. **4D**, upper gate spacers **414** are formed at sidewalls of the gate structure **410**. Cavity spacers **416** are formed in the cavities **412** beneath the upper gate spacers **414**. A deep trench contact etch is then optionally performed to form trenches **418** and to form recessed nanowires **406'**. A patterned relaxed buffer layer **452'** and a patterned defect modification layer **450'** may also be present, as is depicted.

A sacrificial material **420** is then formed in the trenches **418**, as is depicted in FIG. **4E**. In other process schemes, an isolated trench bottom or silicon trench bottom may be used.

Referring to FIG. **4F**, a first epitaxial source or drain structure (e.g., left-hand features **422**) is formed at a first end of the vertical arrangement of horizontal nanowires **406'**. A second epitaxial source or drain structure (e.g., right-hand features **422**) is formed at a second end of the vertical arrangement of horizontal nanowires **406'**. In an embodiment, as depicted, the epitaxial source or drain structures **422** are vertically discrete source or drain structures and may be referred to as epitaxial nubs.

An inter-layer dielectric (ILD) material **424** is then formed at the sides of the gate electrode **410** and adjacent the source or drain structures **422**, as is depicted in FIG. **4G**. Referring to FIG. **4H**, a replacement gate process is used to form a permanent gate dielectric **428** and a permanent gate electrode **426**. The ILD material **424** is then removed, as is depicted in FIG. **4I**. The sacrificial material **420** is then removed from one of the source drain locations (e.g., right-hand side) to form trench **432**, but is not removed from the other of the source drain locations to form trench **430**.

Referring to FIG. **4J**, a first conductive contact structure **434** is formed coupled to the first epitaxial source or drain structure (e.g., left-hand features **422**). A second conductive contact structure **436** is formed coupled to the second epitaxial source or drain structure (e.g., right-hand features **422**). The second conductive contact structure **436** is formed deeper along the fin **402** than the first conductive contact structure **434**. In an embodiment, although not depicted in FIG. **4J**, the method further includes forming an exposed surface of the second conductive contact structure **436** at a bottom of the fin **402**. Conductive contacts may include a contact resistance reducing layer and a primary contact electrode layer, where examples can include Ti, Ni, Co (for the former and W, Ru, Co for the latter.)

In an embodiment, the second conductive contact structure **436** is deeper along the fin **402** than the first conductive contact structure **434**, as is depicted. In one such embodiment, the first conductive contact structure **434** is not along

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the fin **402**, as is depicted. In another such embodiment, not depicted, the first conductive contact structure **434** is partially along the fin **402**.

In an embodiment, the second conductive contact structure **436** is along an entirety of the fin **402**. In an embodiment, although not depicted, in the case that the bottom of the fin **402** is exposed by a back-side substrate removal process, the second conductive contact structure **436** has an exposed surface at a bottom of the fin **402**.

In an embodiment, the structure of FIG. **4J** is included in an integrated circuit structure that also includes a device with channel-to-substrate electrical contact such as described in association with FIG. **2A** or FIG. **2B**. The device with channel-to-substrate electrical contact may be fabricated together with the structure of FIG. **4J** by including process operations that preserve a portion of the sacrificial layers **404** at least in a location beneath a gate electrode.

In another aspect, in order to enable access to both conductive contact structures of a pair of asymmetric source and drain contact structures, integrated circuit structures described herein may be fabricated using a back-side reveal of front-side structures fabrication approach. In some exemplary embodiments, reveal of the back-side of a transistor or other device structure entails wafer-level back-side processing. In contrast to a conventional TSV-type technology, a reveal of the back-side of a transistor as described herein may be performed at the density of the device cells, and even within sub-regions of a device. Furthermore, such a reveal of the back-side of a transistor may be performed to remove substantially all of a donor substrate upon which a device layer was disposed during front-side device processing. As such, a microns-deep TSV becomes unnecessary with the thickness of semiconductor in the device cells following a reveal of the back-side of a transistor potentially being only tens or hundreds of nanometers.

Reveal techniques described herein may enable a paradigm shift from "bottom-up" device fabrication to "center-out" fabrication, where the "center" is any layer that is employed in front-side fabrication, revealed from the back-side, and again employed in back-side fabrication. Processing of both a front-side and revealed back-side of a device structure may address many of the challenges associated with fabricating 3D ICs when primarily relying on front-side processing.

A reveal of the back-side of a transistor approach may be employed for example to remove at least a portion of a carrier layer and intervening layer of a donor-host substrate assembly. The process flow begins with an input of a donor-host substrate assembly. A thickness of a carrier layer in the donor-host substrate is polished (e.g., CMP) and/or etched with a wet or dry (e.g., plasma) etch process. Any grind, polish, and/or wet/dry etch process known to be suitable for the composition of the carrier layer may be employed. For example, where the carrier layer is a group IV semiconductor (e.g., silicon) a CMP slurry known to be suitable for thinning the semiconductor may be employed. Likewise, any wet etchant or plasma etch process known to be suitable for thinning the group IV semiconductor may also be employed.

In some embodiments, the above is preceded by cleaving the carrier layer along a fracture plane substantially parallel to the intervening layer. The cleaving or fracture process may be utilized to remove a substantial portion of the carrier layer as a bulk mass, reducing the polish or etch time needed to remove the carrier layer. For example, where a carrier layer is 400-900 μm in thickness, 100-700 μm may be cleaved off by practicing any blanket implant known to

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promote a wafer-level fracture. In some exemplary embodiments, a light element (e.g., H, He, or Li) is implanted to a uniform target depth within the carrier layer where the fracture plane is desired. Following such a cleaving process, the thickness of the carrier layer remaining in the donor-host substrate assembly may then be polished or etched to complete removal. Alternatively, where the carrier layer is not fractured, the grind, polish and/or etch operation may be employed to remove a greater thickness of the carrier layer.

Next, exposure of an intervening layer is detected. Detection is used to identify a point when the back-side surface of the donor substrate has advanced to nearly the device layer. Any endpoint detection technique known to be suitable for detecting a transition between the materials employed for the carrier layer and the intervening layer may be practiced. In some embodiments, one or more endpoint criteria are based on detecting a change in optical absorbance or emission of the back-side surface of the donor substrate during the polishing or etching performed. In some other embodiments, the endpoint criteria are associated with a change in optical absorbance or emission of byproducts during the polishing or etching of the donor substrate back-side surface. For example, absorbance or emission wavelengths associated with the carrier layer etch byproducts may change as a function of the different compositions of the carrier layer and intervening layer. In other embodiments, the endpoint criteria are associated with a change in mass of species in byproducts of polishing or etching the back-side surface of the donor substrate. For example, the byproducts of processing may be sampled through a quadrupole mass analyzer and a change in the species mass may be correlated to the different compositions of the carrier layer and intervening layer. In another exemplary embodiment, the endpoint criteria is associated with a change in friction between a back-side surface of the donor substrate and a polishing surface in contact with the back-side surface of the donor substrate.

Detection of the intervening layer may be enhanced where the removal process is selective to the carrier layer relative to the intervening layer as non-uniformity in the carrier removal process may be mitigated by an etch rate delta between the carrier layer and intervening layer. Detection may even be skipped if the grind, polish and/or etch operation removes the intervening layer at a rate sufficiently below the rate at which the carrier layer is removed. If an endpoint criteria is not employed, a grind, polish and/or etch operation of a predetermined fixed duration may stop on the intervening layer material if the thickness of the intervening layer is sufficient for the selectivity of the etch. In some examples, the carrier etch rate: intervening layer etch rate is 3:1-10:1, or more.

Upon exposing the intervening layer, at least a portion of the intervening layer may be removed. For example, one or more component layers of the intervening layer may be removed. A thickness of the intervening layer may be removed uniformly by a polish, for example. Alternatively, a thickness of the intervening layer may be removed with a masked or blanket etch process. The process may employ the same polish or etch process as that employed to thin the carrier, or may be a distinct process with distinct process parameters. For example, where the intervening layer provides an etch stop for the carrier removal process, the latter operation may employ a different polish or etch process that favors removal of the intervening layer over removal of the device layer. Where less than a few hundred nanometers of intervening layer thickness is to be removed, the removal process may be relatively slow, optimized for across-wafer

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uniformity, and more precisely controlled than that employed for removal of the carrier layer. A CMP process employed may, for example employ a slurry that offers very high selectivity (e.g., 100:1-300:1, or more) between semiconductor (e.g., silicon) and dielectric material (e.g., SiO) surrounding the device layer and embedded within the intervening layer, for example, as electrical isolation between adjacent device regions.

For embodiments where the device layer is revealed through complete removal of the intervening layer, back-side processing may commence on an exposed back-side of the device layer or specific device regions there in. In some embodiments, the back-side device layer processing includes a further polish or wet/dry etch through a thickness of the device layer disposed between the intervening layer and a device region previously fabricated in the device layer, such as a source or drain region.

In some embodiments where the carrier layer, intervening layer, or device layer back-side is recessed with a wet and/or plasma etch, such an etch may be a patterned etch or a materially selective etch that imparts significant non-planarity or topography into the device layer back-side surface. As described further below, the patterning may be within a device cell (i.e., "intra-cell" patterning) or may be across device cells (i.e., "inter-cell" patterning). In some patterned etch embodiments, at least a partial thickness of the intervening layer is employed as a hard mask for back-side device layer patterning. Hence, a masked etch process may preface a correspondingly masked device layer etch.

The above described processing scheme may result in a donor-host substrate assembly that includes IC devices that have a back-side of an intervening layer, a back-side of the device layer, and/or back-side of one or more semiconductor regions within the device layer, and/or front-side metallization revealed. Additional back-side processing of any of these revealed regions may then be performed during downstream processing.

It is to be appreciated that the structures resulting from the above exemplary processing schemes may be used in a same or similar form for subsequent processing operations to complete device fabrication, such as PMOS and/or NMOS device fabrication. As an example of a completed device, FIG. 5 illustrate a cross-sectional view of a non-planar integrated circuit structure as taken along a gate line, in accordance with an embodiment of the present disclosure.

Referring to FIG. 5, a semiconductor structure or device 500 includes a non-planar active region (e.g., a fin structure including protruding fin portion 504 and sub-fin region 505) within a trench isolation region 506. In an embodiment, instead of a solid fin, the non-planar active region is separated into nanowires (such as nanowires 504A and 504B) above sub-fin region 505, as is represented by the dashed lines. In either case, for ease of description for non-planar integrated circuit structure 500, a non-planar active region 504 is referenced below as a protruding fin portion. In an embodiment, the sub-fin region 505 also includes a relaxed buffer layer 542 and a defect modification layer 540, as is depicted.

A gate line 508 is disposed over the protruding portions 504 of the non-planar active region (including, if applicable, surrounding nanowires 504A and 504B), as well as over a portion of the trench isolation region 506. As shown, gate line 508 includes a gate electrode 550 and a gate dielectric layer 552. In one embodiment, gate line 508 may also include a dielectric cap layer 554. A gate contact 514, and overlying gate contact via 516 are also seen from this perspective, along with an overlying metal interconnect 560,

all of which are disposed in inter-layer dielectric stacks or layers **570**. Also seen from the perspective of FIG. **5**, the gate contact **514** is, in one embodiment, disposed over trench isolation region **506**, but not over the non-planar active regions. In another embodiment, the gate contact **514** is over the non-planar active regions.

In an embodiment, the semiconductor structure or device **500** is a non-planar device such as, but not limited to, a fin-FET device, a tri-gate device, a nanoribbon device, or a nanowire device. In such an embodiment, a corresponding semiconducting channel region is composed of or is formed in a three-dimensional body. In one such embodiment, the gate electrode stacks of gate lines **508** surround at least a top surface and a pair of sidewalls of the three-dimensional body.

As is also depicted in FIG. **5**, in an embodiment, an interface **580** exists between a protruding fin portion **504** and sub-fin region **505**. The interface **580** can be a transition region between a doped sub-fin region **505** and a lightly or undoped upper fin portion **504**. In one such embodiment, each fin is approximately 10 nanometers wide or less, and sub-fin dopants are optionally supplied from an adjacent solid state doping layer at the sub-fin location. In a particular such embodiment, each fin is less than 10 nanometers wide.

Although not depicted in FIG. **5**, it is to be appreciated that source or drain regions of or adjacent to the protruding fin portions **504** are on either side of the gate line **508**, i.e., into and out of the page. In one embodiment, the material of the protruding fin portions **504** in the source or drain locations is removed and replaced with another semiconductor material, e.g., by epitaxial deposition to form epitaxial source or drain structures. The source or drain regions may extend below the height of dielectric layer of trench isolation region **506**, i.e., into the sub-fin region **505**. In accordance with an embodiment of the present disclosure, the more heavily doped sub-fin regions, i.e., the doped portions of the fins below interface **580**, inhibits source to drain leakage through this portion of the bulk semiconductor fins. In an embodiment, the source and drain regions have associated asymmetric source and drain contact structures, as described above in association with FIG. **4J**.

With reference again to FIG. **5**, in an embodiment, fins **504/505** (and, possibly nanowires **504A** and **504B**) are composed of a crystalline silicon germanium layer which may be doped with a charge carrier, such as but not limited to phosphorus, arsenic, boron, gallium or a combination thereof.

In an embodiment, trench isolation region **506**, and trench isolation regions (trench isolation structures or trench isolation layers) described throughout, may be composed of a material suitable to ultimately electrically isolate, or contribute to the isolation of, portions of a permanent gate structure from an underlying bulk substrate or isolate active regions formed within an underlying bulk substrate, such as isolating fin active regions. For example, in one embodiment, trench isolation region **506** is composed of a dielectric material such as, but not limited to, silicon dioxide, silicon oxy-nitride, silicon nitride, or carbon-doped silicon nitride.

Gate line **508** may be composed of a gate electrode stack which includes a gate dielectric layer **552** and a gate electrode layer **550**. In an embodiment, the gate electrode of the gate electrode stack is composed of a metal gate and the gate dielectric layer is composed of a high-k material. For example, in one embodiment, the gate dielectric layer **552** is composed of a material such as, but not limited to, hafnium oxide, hafnium oxy-nitride, hafnium silicate, lanthanum oxide, zirconium oxide, zirconium silicate, tantalum oxide,

barium strontium titanate, barium titanate, strontium titanate, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, lead zinc niobate, or a combination thereof. Furthermore, a portion of gate dielectric layer **552** may include a layer of native oxide formed from the top few layers of the substrate fin **504**. In an embodiment, the gate dielectric layer **552** is composed of a top high-k portion and a lower portion composed of an oxide of a semiconductor material. In one embodiment, the gate dielectric layer **552** is composed of a top portion of hafnium oxide and a bottom portion of silicon dioxide or silicon oxy-nitride. In some implementations, a portion of the gate dielectric is a "U"-shaped structure that includes a bottom portion substantially parallel to the surface of the substrate and two sidewall portions that are substantially perpendicular to the top surface of the substrate.

In one embodiment, the gate electrode layer **550** is composed of a metal layer such as, but not limited to, metal nitrides, metal carbides, metal silicides, metal aluminides, hafnium, zirconium, titanium, tantalum, aluminum, ruthenium, palladium, platinum, cobalt, nickel or conductive metal oxides. In a specific embodiment, the gate electrode layer **550** is composed of a non-workfunction-setting fill material formed above a metal workfunction-setting layer. The gate electrode layer **550** may consist of a P-type workfunction metal or an N-type workfunction metal, depending on whether the transistor is to be a PMOS or an NMOS transistor. In some implementations, the gate electrode layer **550** may consist of a stack of two or more metal layers, where one or more metal layers are workfunction metal layers and at least one metal layer is a conductive fill layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, and conductive metal oxides, e.g., ruthenium oxide. A P-type metal layer will enable the formation of a PMOS gate electrode with a workfunction that is between about 4.9 eV and about 5.2 eV. For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, and carbides of these metals such as hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide. An N-type metal layer will enable the formation of an NMOS gate electrode with a workfunction that is between about 3.9 eV and about 4.2 eV. In some implementations, the gate electrode may consist of a "U"-shaped structure that includes a bottom portion substantially parallel to the surface of the substrate and two sidewall portions that are substantially perpendicular to the top surface of the substrate. In another implementation, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the substrate and does not include sidewall portions substantially perpendicular to the top surface of the substrate. In further implementations of the disclosure, the gate electrode may consist of a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may consist of one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

Spacers associated with the gate electrode stacks may be composed of a material suitable to ultimately electrically isolate, or contribute to the isolation of, a permanent gate structure from adjacent conductive contacts, such as self-aligned contacts. For example, in one embodiment, the spacers are composed of a dielectric material such as, but not

limited to, silicon dioxide, silicon oxy-nitride, silicon nitride, or carbon-doped silicon nitride.

Gate contact **514** and overlying gate contact via **516** may be composed of a conductive material. In an embodiment, one or more of the contacts or vias are composed of a metal species. The metal species may be a pure metal, such as tungsten, nickel, or cobalt, or may be an alloy such as a metal-metal alloy or a metal-semiconductor alloy (e.g., such as a silicide material).

In an embodiment (although not shown), a contact pattern which is essentially perfectly aligned to an existing gate pattern **508** is formed while eliminating the use of a lithographic step with exceedingly tight registration budget. In an embodiment, the contact pattern is a vertically symmetric contact pattern, or an asymmetric contact pattern such as described in association with FIG. 4J. In other embodiments, all contacts are front-side connected and are not asymmetric. In one such embodiment, the self-aligned approach enables the use of intrinsically highly selective wet etching (e.g., versus conventionally implemented dry or plasma etching) to generate contact openings. In an embodiment, a contact pattern is formed by utilizing an existing gate pattern in combination with a contact plug lithography operation. In one such embodiment, the approach enables elimination of the need for an otherwise critical lithography operation to generate a contact pattern, as used in conventional approaches. In an embodiment, a trench contact grid is not separately patterned, but is rather formed between poly (gate) lines. For example, in one such embodiment, a trench contact grid is formed subsequent to gate grating patterning but prior to gate grating cuts.

In an embodiment, providing structure **500** involves fabrication of the gate stack structure **508** by a replacement gate process. In such a scheme, dummy gate material such as polysilicon or silicon nitride pillar material, may be removed and replaced with permanent gate electrode material. In one such embodiment, a permanent gate dielectric layer is also formed in this process, as opposed to being carried through from earlier processing. In an embodiment, dummy gates are removed by a dry etch or wet etch process. In one embodiment, dummy gates are composed of polycrystalline silicon or amorphous silicon and are removed with a dry etch process including use of SF_6 . In another embodiment, dummy gates are composed of polycrystalline silicon or amorphous silicon and are removed with a wet etch process including use of aqueous NH_4OH or tetramethylammonium hydroxide. In one embodiment, dummy gates are composed of silicon nitride and are removed with a wet etch including aqueous phosphoric acid.

Referring again to FIG. 5, the arrangement of semiconductor structure or device **500** places the gate contact over isolation regions. Such an arrangement may be viewed as inefficient use of layout space. In another embodiment, however, a semiconductor device has contact structures that contact portions of a gate electrode formed over an active region, e.g., over a fin **505**, and in a same layer as a trench contact via.

In an embodiment, the structure of FIG. 5 is included in an integrated circuit structure that also includes a device with channel-to-substrate electrical contact such as described in association with FIG. 2A or FIG. 2B. The device with channel-to-substrate electrical contact may be fabricated together with the structure of FIG. 5 by including process operations that preserve a portion of a sacrificial layer between and below nanowires **504A** and **504B** at least in a location beneath a gate electrode.

It is to be appreciated that not all aspects of the processes described above need be practiced to fall within the spirit and scope of embodiments of the present disclosure. Also, the processes described herein may be used to fabricate one or a plurality of semiconductor devices. The semiconductor devices may be transistors or like devices. For example, in an embodiment, the semiconductor devices are a metal-oxide semiconductor (MOS) transistors for logic or memory, or are bipolar transistors. Also, in an embodiment, the semiconductor devices have a three-dimensional architecture, such as a tri-gate device, an independently accessed double gate device, or a FIN-FET. One or more embodiments may be particularly useful for fabricating semiconductor devices at a sub-10 nanometer (10 nm) technology node.

In an embodiment, as used throughout the present description, interlayer dielectric (ILD) material is composed of or includes a layer of a dielectric or insulating material. Examples of suitable dielectric materials include, but are not limited to, oxides of silicon (e.g., silicon dioxide (SiO_2)), doped oxides of silicon, fluorinated oxides of silicon, carbon doped oxides of silicon, various low-k dielectric materials known in the arts, and combinations thereof. The interlayer dielectric material may be formed by conventional techniques, such as, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), or by other deposition methods.

In an embodiment, as is also used throughout the present description, metal lines or interconnect line material (and via material) is composed of one or more metal or other conductive structures. A common example is the use of copper lines and structures that may or may not include barrier layers between the copper and surrounding ILD material. As used herein, the term metal includes alloys, stacks, and other combinations of multiple metals. For example, the metal interconnect lines may include barrier layers (e.g., layers including one or more of Ta, TaN, Ti or TiN), stacks of different metals or alloys, etc. Thus, the interconnect lines may be a single material layer, or may be formed from several layers, including conductive liner layers and fill layers. Any suitable deposition process, such as electroplating, chemical vapor deposition or physical vapor deposition, may be used to form interconnect lines. In an embodiment, the interconnect lines are composed of a conductive material such as, but not limited to, Cu, Al, Ti, Zr, Hf, V, Ru, Co, Ni, Pd, Pt, W, Ag, Au or alloys thereof. The interconnect lines are also sometimes referred to in the art as traces, wires, lines, metal, or simply interconnect.

In an embodiment, as is also used throughout the present description, hardmask materials, capping layers, or plugs are composed of dielectric materials different from the interlayer dielectric material. In one embodiment, different hardmask, capping or plug materials may be used in different regions so as to provide different growth or etch selectivity to each other and to the underlying dielectric and metal layers. In some embodiments, a hardmask layer, capping or plug layer includes a layer of a nitride of silicon (e.g., silicon nitride) or a layer of an oxide of silicon, or both, or a combination thereof. Other suitable materials may include carbon-based materials. Other hardmask, capping or plug layers known in the arts may be used depending upon the particular implementation. The hardmask, capping or plug layers may be formed by CVD, PVD, or by other deposition methods.

In an embodiment, as is also used throughout the present description, lithographic operations are performed using 193 nm immersion lithography (i193), EUV and/or EBDW

lithography, or the like. A positive tone or a negative tone resist may be used. In one embodiment, a lithographic mask is a trilayer mask composed of a topographic masking portion, an anti-reflective coating (ARC) layer, and a photoresist layer. In a particular such embodiment, the topographic masking portion is a carbon hardmask (CHM) layer and the anti-reflective coating layer is a silicon ARC layer.

In another aspect, one or more embodiments are directed to neighboring semiconductor structures or devices separated by self-aligned gate endcap (SAGE) structures. Particular embodiments may be directed to integration of multiple width (multi-Wsi) nanowires and nanoribbons in a SAGE architecture and separated by a SAGE wall. In an embodiment, nanowires/nanoribbons are integrated with multiple Wsi in a SAGE architecture portion of a front-end process flow. Such a process flow may involve integration of nanowires and nanoribbons of different Wsi to provide robust functionality of next generation transistors with low power and high performance. Associated epitaxial source or drain regions may be embedded (e.g., portions of nanowires removed) and then source or drain (S/D) growth is performed).

To provide further context, advantages of a self-aligned gate endcap (SAGE) architecture may include the enabling of higher layout density and, in particular, scaling of diffusion to diffusion spacing. To provide illustrative comparison, FIG. 6 illustrates cross-sectional views taken through nanowires and fins for a non-endcap architecture (left-hand side (a)) versus a self-aligned gate endcap (SAGE) architecture (right-hand side (b)), in accordance with an embodiment of the present disclosure.

Referring to the left-hand side (a) of FIG. 6, an integrated circuit structure 600 includes a substrate 602 having fins 604 protruding there from by an amount 606 above an isolation structure 608 laterally surrounding lower portions of the fins 604. Upper portions of the fins may include a relaxed buffer layer 622 and a defect modification layer 620, as is depicted. Corresponding nanowires 605 are over the fins 604. A gate structure may be formed over the integrated circuit structure 600 to fabricate a device. However, breaks in such a gate structure may be accommodated for by increasing the spacing between fin 604/nanowire 605 pairs.

By contrast, referring to the right-hand side (b) of FIG. 6, an integrated circuit structure 650 includes a substrate 652 having fins 654 protruding therefrom by an amount 656 above an isolation structure 658 laterally surrounding lower portions of the fins 654. Upper portions of the fins may include a relaxed buffer layer 672 and a defect modification layer 670, as is depicted. Corresponding nanowires 655 are over the fins 654. Isolating SAGE walls 660 (which may include a hardmask thereon, as depicted) are included within the isolation structure 652 and between adjacent fin 654/nanowire 655 pairs. The distance between an isolating SAGE wall 660 and a nearest fin 654/nanowire 655 pair defines the gate endcap spacing 662. A gate structure may be formed over the integrated circuit structure 600, between isolating SAGE walls to fabricate a device. Breaks in such a gate structure are imposed by the isolating SAGE walls. Since the isolating SAGE walls 660 are self-aligned, restrictions from conventional approaches can be minimized to enable more aggressive diffusion to diffusion spacing. Furthermore, since gate structures include breaks at all locations, individual gate structure portions may be layer connected by local interconnects formed over the isolating SAGE walls 660. In an embodiment, as depicted, the SAGE walls 660 each include a lower dielectric portion and a dielectric cap on the lower dielectric portion. In accordance

with an embodiment of the present disclosure, a fabrication process for structures associated with FIG. 6 involves use of a process scheme that provides a gate-all-around integrated circuit structure having epitaxial source or drain structures.

In an embodiment, the structure of part (b) of FIG. 6 is included in an integrated circuit structure that also includes a device with channel-to-substrate electrical contact such as described in association with FIG. 2A or FIG. 2B. The device with channel-to-substrate electrical contact may be fabricated together with the structure of part (b) of FIG. 6 by including process operations that preserve a portion of a sacrificial layer between and below nanowires 655 at least in a location beneath a gate electrode.

A self-aligned gate endcap (SAGE) processing scheme involves the formation of gate/trench contact endcaps self-aligned to fins without requiring an extra length to account for mask mis-registration. Thus, embodiments may be implemented to enable shrinking of transistor layout area. Embodiments described herein may involve the fabrication of gate endcap isolation structures, which may also be referred to as gate walls, isolation gate walls or self-aligned gate endcap (SAGE) walls.

In an exemplary processing scheme for structures having SAGE walls separating neighboring devices, FIG. 7 illustrate cross-sectional views representing various operations in a method of fabricating a self-aligned gate endcap (SAGE) structure with gate-all-around devices, in accordance with an embodiment of the present disclosure.

Referring to part (a) of FIG. 7, a starting structure includes a nanowire patterning stack 704 above a substrate 702. A lithographic patterning stack 706 is formed above the nanowire patterning stack 704. The nanowire patterning stack 704 includes alternating sacrificial layers 710 and nanowire layers 712, which may be above a relaxed buffer layer 782 and a defect modification layer 780, as is depicted. A protective mask 714 is between the nanowire patterning stack 704 and the lithographic patterning stack 706. In one embodiment, the lithographic patterning stack 706 is trilayer mask composed of a topographic masking portion 720, an anti-reflective coating (ARC) layer 722, and a photoresist layer 724. In a particular such embodiment, the topographic masking portion 720 is a carbon hardmask (CHM) layer and the anti-reflective coating layer 722 is a silicon ARC layer.

Referring to part (b) of FIG. 7, the stack of part (a) is lithographically patterned and then etched to provide an etched structure including a patterned substrate 702 and trenches 730.

Referring to part (c) of FIG. 7, the structure of part (b) has an isolation layer 740 and a SAGE material 742 formed in trenches 730. The structure is then planarized to leave patterned topographic masking layer 720' as an exposed upper layer.

Referring to part (d) of FIG. 7, the isolation layer 740 is recessed below an upper surface of the patterned substrate 702, e.g., to define a protruding fin portion and to provide a trench isolation structure 741 beneath SAGE walls 742.

Referring to part (e) of FIG. 7, the sacrificial layers 710 are removed at least in the channel region to release nanowires 712A and 712B. Subsequent to the formation of the structure of part (e) of FIG. 7, a gate stacks may be formed around nanowires 712B or 712A, over protruding fins of substrate 702, and between SAGE walls 742. In one embodiment, prior to formation of the gate stacks, the remaining portion of protective mask 714 is removed. In another embodiment, the remaining portion of protective mask 714 is retained as an insulating fin hat as an artifact of the processing scheme.

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Referring again to part (e) of FIG. 7, it is to be appreciated that a channel view is depicted, with source or drain regions being locating into and out of the page. In an embodiment, the channel region including nanowires **712B** has a width less than the channel region including nanowires **712A**. Thus, in an embodiment, an integrated circuit structure includes multiple width (multi-Wsi) nanowires. Although structures of **712B** and **712A** may be differentiated as nanowires and nanoribbons, respectively, both such structures are typically referred to herein as nanowires. It is also to be appreciated that reference to or depiction of a fin/nanowire pair throughout may refer to a structure including a fin and one or more overlying nanowires (e.g., two overlying nanowires are shown in FIG. 7). In accordance with an embodiment of the present disclosure, a fabrication process for structures associated with FIG. 7 involves use of a process scheme that provides a gate-all-around integrated circuit structure having epitaxial source or drain structures.

In an embodiment, the structure of part (e) FIG. 7 is included in an integrated circuit structure that also includes a device with channel-to-substrate electrical contact such as described in association with FIG. 2A or FIG. 2B. The device with channel-to-substrate electrical contact may be fabricated together with the structure of part (e) of FIG. 7 by including process operations that preserve a portion of a sacrificial layer between and below nanowires **712A** and/or nanowires or ribbons **712B** at least in a location beneath a gate electrode.

In an embodiment, as described throughout, self-aligned gate endcap (SAGE) isolation structures may be composed of a material or materials suitable to ultimately electrically isolate, or contribute to the isolation of, portions of permanent gate structures from one another. Exemplary materials or material combinations include a single material structure such as silicon dioxide, silicon oxy-nitride, silicon nitride, or carbon-doped silicon nitride. Other exemplary materials or material combinations include a multi-layer stack having lower portion silicon dioxide, silicon oxy-nitride, silicon nitride, or carbon-doped silicon nitride and an upper portion higher dielectric constant material such as hafnium oxide.

To highlight an exemplary integrated circuit structure having three vertically arranged nanowires, FIG. 8A illustrates a three-dimensional cross-sectional view of a nanowire-based integrated circuit structure, in accordance with an embodiment of the present disclosure. FIG. 8B illustrates a cross-sectional source or drain view of the nanowire-based integrated circuit structure of FIG. 8A, as taken along the a-a' axis. FIG. 8C illustrates a cross-sectional channel view of the nanowire-based integrated circuit structure of FIG. 8A, as taken along the b-b' axis.

Referring to FIG. 8A, an integrated circuit structure **800** includes one or more vertically stacked nanowires (**804** set) above a substrate **802**. In an embodiment, as depicted, a relaxed buffer layer **802C**, a defect modification layer **802B**, and a lower substrate portion **802A** are included in substrate **802**, as is depicted. An optional fin below the bottommost nanowire and formed from the substrate **802** is not depicted for the sake of emphasizing the nanowire portion for illustrative purposes. Embodiments herein are targeted at both single wire devices and multiple wire devices. As an example, a three nanowire-based devices having nanowires **804A**, **804B** and **804C** is shown for illustrative purposes. For convenience of description, nanowire **804A** is used as an example where description is focused on one of the nanowires. It is to be appreciated that where attributes of one nanowire are described, embodiments based on a plurality of

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nanowires may have the same or essentially the same attributes for each of the nanowires.

Each of the nanowires **804** includes a channel region **806** in the nanowire. The channel region **806** has a length (L). Referring to FIG. 8C, the channel region also has a perimeter (Pc) orthogonal to the length (L). Referring to both FIGS. 8A and 8C, a gate electrode stack **808** surrounds the entire perimeter (Pc) of each of the channel regions **806**. The gate electrode stack **808** includes a gate electrode along with a gate dielectric layer between the channel region **806** and the gate electrode (not shown). In an embodiment, the channel region is discrete in that it is completely surrounded by the gate electrode stack **808** without any intervening material such as underlying substrate material or overlying channel fabrication materials. Accordingly, in embodiments having a plurality of nanowires **804**, the channel regions **806** of the nanowires are also discrete relative to one another.

Referring to both FIGS. 8A and 8B, integrated circuit structure **800** includes a pair of non-discrete source or drain regions **810/812**. The pair of non-discrete source or drain regions **810/812** is on either side of the channel regions **806** of the plurality of vertically stacked nanowires **804**. Furthermore, the pair of non-discrete source or drain regions **810/812** is adjoining for the channel regions **806** of the plurality of vertically stacked nanowires **804**. In one such embodiment, not depicted, the pair of non-discrete source or drain regions **810/812** is directly vertically adjoining for the channel regions **806** in that epitaxial growth is on and between nanowire portions extending beyond the channel regions **806**, where nanowire ends are shown within the source or drain structures. In another embodiment, as depicted in FIG. 8A, the pair of non-discrete source or drain regions **810/812** is indirectly vertically adjoining for the channel regions **806** in that they are formed at the ends of the nanowires and not between the nanowires.

In an embodiment, as depicted, the source or drain regions **810/812** are non-discrete in that there are not individual and discrete source or drain regions for each channel region **806** of a nanowire **804**. Accordingly, in embodiments having a plurality of nanowires **804**, the source or drain regions **810/812** of the nanowires are global or unified source or drain regions as opposed to discrete for each nanowire. That is, the non-discrete source or drain regions **810/812** are global in the sense that a single unified feature is used as a source or drain region for a plurality (in this case, 3) of nanowires **804** and, more particularly, for more than one discrete channel region **806**. In one embodiment, from a cross-sectional perspective orthogonal to the length of the discrete channel regions **806**, each of the pair of non-discrete source or drain regions **810/812** is approximately rectangular in shape with a bottom tapered portion and a top vertex portion, as depicted in FIG. 8B. In other embodiments, however, the source or drain regions **810/812** of the nanowires are relatively larger yet discrete non-vertically merged epitaxial structures such as nubs described in association with FIGS. 4A-4J.

In accordance with an embodiment of the present disclosure, and as depicted in FIGS. 8A and 8B, integrated circuit structure **800** further includes a pair of contacts **814**, each contact **814** on one of the pair of non-discrete source or drain regions **810/812**. In one such embodiment, in a vertical sense, each contact **814** completely surrounds the respective non-discrete source or drain region **810/812**. In another aspect, the entire perimeter of the non-discrete source or drain regions **810/812** may not be accessible for contact with contacts **814**, and the contact **814** thus only partially surrounds the non-discrete source or drain regions **810/812**, as

depicted in FIG. 8B. In a contrasting embodiment, not depicted, the entire perimeter of the non-discrete source or drain regions **810/812**, as taken along the a-a' axis, is surrounded by the contacts **814**.

Referring again to FIG. 8A, in an embodiment, integrated circuit structure **800** further includes a pair of spacers **816**. As is depicted, outer portions of the pair of spacers **816** may overlap portions of the non-discrete source or drain regions **810/812**, providing for "embedded" portions of the non-discrete source or drain regions **810/812** beneath the pair of spacers **816**. As is also depicted, the embedded portions of the non-discrete source or drain regions **810/812** may not extend beneath the entirety of the pair of spacers **816**.

Substrate **802** may be composed of a material suitable for integrated circuit structure fabrication. In one embodiment, substrate **802** includes a lower bulk substrate composed of a single crystal of a material which may include, but is not limited to, silicon, germanium, silicon-germanium, germanium-tin, silicon-germanium-tin, or a group III-V compound semiconductor material. An upper insulator layer composed of a material which may include, but is not limited to, silicon dioxide, silicon nitride or silicon oxy-nitride is on the lower bulk substrate. Thus, the structure **800** may be fabricated from a starting semiconductor-on-insulator substrate. Alternatively, the structure **800** is formed directly from a bulk substrate and local oxidation is used to form electrically insulative portions in place of the above described upper insulator layer. In another alternative embodiment, the structure **800** is formed directly from a bulk substrate and doping is used to form electrically isolated active regions, such as nanowires, thereon. In one such embodiment, the first nanowire (i.e., proximate the substrate) is in the form of an omega-FET type structure.

In an embodiment, the nanowires **804** may be sized as wires or ribbons, as described below, and may have squared-off or rounder corners. In an embodiment, the nanowires **804** are composed of a material such as, but not limited to, silicon, germanium, or a combination thereof. In one such embodiment, the nanowires are single-crystalline. For example, for a silicon nanowire **804**, a single-crystalline nanowire may be based from a (100) global orientation, e.g., with a <100> plane in the z-direction. As described below, other orientations may also be considered. In an embodiment, the dimensions of the nanowires **804**, from a cross-sectional perspective, are on the nano-scale. For example, in a specific embodiment, the smallest dimension of the nanowires **804** is less than approximately 20 nanometers. In an embodiment, the nanowires **804** are composed of a strained material, particularly in the channel regions **806**.

Referring to FIGS. 8C, in an embodiment, each of the channel regions **806** has a width (Wc) and a height (Hc), the width (Wc) approximately the same as the height (Hc). That is, in both cases, the channel regions **806** are square-like or, if corner-rounded, circle-like in cross-section profile. In another aspect, the width and height of the channel region need not be the same, such as the case for nanoribbons as described throughout.

In an embodiment, as described throughout, an integrated circuit structure includes non-planar devices such as, but not limited to, a finFET or a tri-gate device with corresponding one or more overlying nanowire structures. In such an embodiment, a corresponding semiconducting channel region is composed of or is formed in a three-dimensional body with one or more discrete nanowire channel portions overlying the three-dimensional body. In one such embodiment, the gate structures surround at least a top surface and

a pair of sidewalls of the three-dimensional body, and further surrounds each of the one or more discrete nanowire channel portions.

In an embodiment, the structure of FIGS. **8A-8C** is included in an integrated circuit structure that also includes a device with channel-to-substrate electrical contact such as described in association with FIG. 2A or FIG. 2B. The device with channel-to-substrate electrical contact may be fabricated together with the structure of FIGS. **8A-8C** by including process operations that preserve a portion of a sacrificial layer between and below nanowires **804** at least in a location beneath a gate electrode.

In an embodiment, as described throughout, an underlying substrate may be composed of a semiconductor material that can withstand a manufacturing process and in which charge can migrate. In an embodiment, the substrate is a bulk substrate composed of a crystalline silicon, silicon/germanium or germanium layer doped with a charge carrier, such as but not limited to phosphorus, arsenic, boron, gallium or a combination thereof, to form an active region. In one embodiment, the concentration of silicon atoms in a bulk substrate is greater than 97%. In another embodiment, a bulk substrate is composed of an epitaxial layer grown atop a distinct crystalline substrate, e.g. a silicon epitaxial layer grown atop a boron-doped bulk silicon mono-crystalline substrate. A bulk substrate may alternatively be composed of a group III-V material. In an embodiment, a bulk substrate is composed of a group III-V material such as, but not limited to, gallium nitride, gallium phosphide, gallium arsenide, indium phosphide, indium antimonide, indium gallium arsenide, aluminum gallium arsenide, indium gallium phosphide, or a combination thereof. In one embodiment, a bulk substrate is composed of a group III-V material and the charge-carrier dopant impurity atoms are ones such as, but not limited to, carbon, silicon, germanium, oxygen, sulfur, selenium or tellurium.

Embodiments disclosed herein may be used to manufacture a wide variety of different types of integrated circuits and/or microelectronic devices. Examples of such integrated circuits include, but are not limited to, processors, chipset components, graphics processors, digital signal processors, micro-controllers, and the like. In other embodiments, semiconductor memory may be manufactured. Moreover, the integrated circuits or other microelectronic devices may be used in a wide variety of electronic devices known in the arts. For example, in computer systems (e.g., desktop, laptop, server), cellular phones, personal electronics, etc. The integrated circuits may be coupled with a bus and other components in the systems. For example, a processor may be coupled by one or more buses to a memory, a chipset, etc. Each of the processor, the memory, and the chipset, may potentially be manufactured using the approaches disclosed herein.

FIG. 9 illustrates a computing device **900** in accordance with one implementation of an embodiment of the present disclosure. The computing device **900** houses a board **902**. The board **902** may include a number of components, including but not limited to a processor **904** and at least one communication chip **906**. The processor **904** is physically and electrically coupled to the board **902**. In some implementations the at least one communication chip **906** is also physically and electrically coupled to the board **902**. In further implementations, the communication chip **906** is part of the processor **904**.

Depending on its applications, computing device **900** may include other components that may or may not be physically and electrically coupled to the board **902**. These other

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components include, but are not limited to, volatile memory (e.g., DRAM), non-volatile memory (e.g., ROM), flash memory, a graphics processor, a digital signal processor, a crypto processor, a chipset, an antenna, a display, a touchscreen display, a touchscreen controller, a battery, an audio codec, a video codec, a power amplifier, a global positioning system (GPS) device, a compass, an accelerometer, a gyroscope, a speaker, a camera, and a mass storage device (such as hard disk drive, compact disk (CD), digital versatile disk (DVD), and so forth).

The communication chip **906** enables wireless communications for the transfer of data to and from the computing device **900**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. The communication chip **906** may implement any of a number of wireless standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The computing device **900** may include a plurality of communication chips **906**. For instance, a first communication chip **906** may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip **906** may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

The processor **904** of the computing device **900** includes an integrated circuit die packaged within the processor **904**. The integrated circuit die of the processor **904** may include one or more structures, such as gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact, built in accordance with implementations of embodiments of the present disclosure. The term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory.

The communication chip **906** also includes an integrated circuit die packaged within the communication chip **906**. The integrated circuit die of the communication chip **906** may include one or more structures, such as gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact, built in accordance with implementations of embodiments of the present disclosure.

In further implementations, another component housed within the computing device **900** may contain an integrated circuit die that includes one or structures, such as gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact, built in accordance with implementations of embodiments of the present disclosure.

In various implementations, the computing device **900** may be a laptop, a netbook, a notebook, an ultrabook, a smartphone, a tablet, a personal digital assistant (PDA), an ultra mobile PC, a mobile phone, a desktop computer, a server, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a digital camera, a portable music player, or a digital video recorder. In further implementations, the computing device **900** may be any other electronic device that processes data.

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FIG. **10** illustrates an interposer **1000** that includes one or more embodiments of the present disclosure. The interposer **1000** is an intervening substrate used to bridge a first substrate **1002** to a second substrate **1004**. The first substrate **1002** may be, for instance, an integrated circuit die. The second substrate **1004** may be, for instance, a memory module, a computer motherboard, or another integrated circuit die. Generally, the purpose of an interposer **1000** is to spread a connection to a wider pitch or to reroute a connection to a different connection. For example, an interposer **1000** may couple an integrated circuit die to a ball grid array (BGA) **1006** that can subsequently be coupled to the second substrate **1004**. In some embodiments, the first and second substrates **1002/1004** are attached to opposing sides of the interposer **1000**. In other embodiments, the first and second substrates **1002/1004** are attached to the same side of the interposer **1000**. And in further embodiments, three or more substrates are interconnected by way of the interposer **1000**.

The interposer **1000** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, a ceramic material, or a polymer material such as polyimide. In further implementations, the interposer **1000** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials.

The interposer **1000** may include metal interconnects **1008** and vias **1010**, including but not limited to through-silicon vias (TSVs) **1012**. The interposer **1000** may further include embedded devices **1014**, including both passive and active devices. Such devices include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, and electrostatic discharge (ESD) devices. More complex devices such as radio-frequency (RF) devices, power amplifiers, power management devices, antennas, arrays, sensors, and MEMS devices may also be formed on the interposer **1000**. In accordance with embodiments of the disclosure, apparatuses or processes disclosed herein may be used in the fabrication of interposer **1000** or in the fabrication of components included in the interposer **1000**.

Thus, embodiments of the present disclosure include gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact, and methods of fabricating gate-all-around integrated circuit structures having devices with channel-to-substrate electrical contact.

The above description of illustrated implementations of embodiments of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize.

These modifications may be made to the disclosure in light of the above detailed description. The terms used in the following claims should not be construed to limit the disclosure to the specific implementations disclosed in the specification and the claims. Rather, the scope of the disclosure is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

Example embodiment 1: An integrated circuit structure includes a first vertical arrangement of horizontal nanowires above a first fin. A channel region of the first vertical arrangement of horizontal nanowires is electrically coupled to the first fin by a semiconductor material layer directly

between the first vertical arrangement of horizontal nanowires and the first fin. A first gate stack is over the first vertical arrangement of horizontal nanowires. A second vertical arrangement of horizontal nanowires is above a second fin. A channel region of the second vertical arrangement of horizontal nanowires is electrically isolated from the second fin. A second gate stack is over the second vertical arrangement of horizontal nanowires.

Example embodiment 2: The integrated circuit structure of example embodiment 1, wherein the semiconductor material layer is between the channel region of the first vertical arrangement of horizontal nanowires and the first fin, and between source or drain regions of the first vertical arrangement of horizontal nanowires and the first fin.

Example embodiment 3: The integrated circuit structure of example embodiment 1 or 2, further including a pair of epitaxial source or drain structures at first and second ends of the second vertical arrangement of horizontal nanowires.

Example embodiment 4: The integrated circuit structure of example embodiment 3, wherein the pair of epitaxial source or drain structures is a pair of non-discrete epitaxial source or drain structures.

Example embodiment 5: The integrated circuit structure of example embodiment 3, wherein the pair of epitaxial source or drain structures is a pair of discrete epitaxial source or drain structures.

Example embodiment 6: The integrated circuit structure of example embodiment 1, 2, 3, 4 or 5, wherein there are no epitaxial source or drain structures at first and second ends of the first vertical arrangement of horizontal nanowires.

Example embodiment 7: The integrated circuit structure of example embodiment 1, 2, 3, 4, 5 or 6, wherein the first and second fins each include a portion of a bulk silicon substrate.

Example embodiment 8: The integrated circuit structure of example embodiment 1, 2, 3, 4, 5, 6 or 7, wherein the nanowires of the first and second vertical arrangements of horizontal nanowires include silicon, and wherein the semiconductor material layer includes silicon and germanium.

Example embodiment 9: The integrated circuit structure of example embodiment 1, 2, 3, 4, 5, 6, 7 or 8, wherein each of the first and second gate stacks include a high-k gate dielectric layer and a metal gate electrode.

Example embodiment 10: An integrated circuit structure includes a first vertical arrangement of horizontal nanowires above a first fin. A channel region of the first vertical arrangement of horizontal nanowires is electrically coupled to the first fin by a semiconductor material layer directly between the channel region of the first vertical arrangement of horizontal nanowires and the first fin but not between source or drain regions of the first vertical arrangement of horizontal nanowires and the first fin. A first gate stack is over the first vertical arrangement of horizontal nanowires. A second vertical arrangement of horizontal nanowires is above a second fin. A channel region of the second vertical arrangement of horizontal nanowires is electrically isolated from the second fin. A second gate stack is over the second vertical arrangement of horizontal nanowires.

Example embodiment 11: The integrated circuit structure of example embodiment 10, further including a first pair of epitaxial source or drain structures at first and second ends of the first vertical arrangement of horizontal nanowires, and a second pair of epitaxial source or drain structures at first and second ends of the second vertical arrangement of horizontal nanowires.

Example embodiment 12: The integrated circuit structure of example embodiment 11, wherein the first and second

pairs of epitaxial source or drain structures are first and second pairs of non-discrete epitaxial source or drain structures.

Example embodiment 13: The integrated circuit structure of example embodiment 11, wherein the first and second pairs of epitaxial source or drain structures are first and second pairs of discrete epitaxial source or drain structures.

Example embodiment 14: The integrated circuit structure of example embodiment 10, 11, 12 or 13, wherein the first and second fins each include a portion of a bulk silicon substrate.

Example embodiment 15: The integrated circuit structure of example embodiment 10, 11, 12, 13 or 14, wherein the nanowires of the first and second vertical arrangements of horizontal nanowires include silicon, and wherein the semiconductor material layer includes silicon and germanium.

Example embodiment 16: The integrated circuit structure of example embodiment 10, 11, 12, 13, 14 or 15, wherein each of the first and second gate stacks include a high-k gate dielectric layer and a metal gate electrode.

Example embodiment 17: A computing device includes a board, and a component coupled to the board. The component includes an integrated circuit structure including a first vertical arrangement of horizontal nanowires above a first fin. A channel region of the first vertical arrangement of horizontal nanowires is electrically coupled to the first fin by a semiconductor material layer directly between the first vertical arrangement of horizontal nanowires and the first fin. A first gate stack is over the first vertical arrangement of horizontal nanowires. A second vertical arrangement of horizontal nanowires is above a second fin. A channel region of the second vertical arrangement of horizontal nanowires is electrically isolated from the second fin. A second gate stack is over the second vertical arrangement of horizontal nanowires.

Example embodiment 18: The computing device of example embodiment 17, further including a memory coupled to the board.

Example embodiment 19: The computing device of example embodiment 17 or 18, further including a communication chip coupled to the board.

Example embodiment 20: The computing device of example embodiment 17, 18 or 19, wherein the component is a packaged integrated circuit die.

Example embodiment 21: The computing device of example embodiment 17, 18, 19 or 20, wherein the component is selected from the group consisting of a processor, a communications chip, and a digital signal processor.

Example embodiment 22: The computing device of example embodiment 17, 18, 19, 20 or 21, wherein the computing device is selected from the group consisting of a mobile phone, a laptop, a desk top computer, a server, and a set-top box.

What is claimed is:

1. An integrated circuit structure, comprising:
 - a first nanowire above a first fin, a channel region of the first nanowire electrically coupled to the first fin by a semiconductor material layer directly between the first nanowire and the first fin;
 - a first gate stack over the first nanowire;
 - a second nanowire above a second fin, a channel region of the second nanowire electrically isolated from the second fin;
 - a second gate stack over the second nanowire; and
 - a pair of epitaxial source or drain structures at first and second ends of the second nanowire, wherein the pair

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of epitaxial source or drain structures is vertically separated from the second fin.

2. The integrated circuit structure of claim 1, wherein the semiconductor material layer is between the channel region of the first nanowire and the first fin, and between source or drain regions of the first nanowire and the first fin.

3. The integrated circuit structure of claim 1, wherein the pair of epitaxial source or drain structures is a pair of non-discrete epitaxial source or drain structures.

4. The integrated circuit structure of claim 1, wherein the pair of epitaxial source or drain structures is a pair of discrete epitaxial source or drain structures.

5. The integrated circuit structure of claim 1, wherein there are no epitaxial source or drain structures at first and second ends of the first nanowire.

6. The integrated circuit structure of claim 1, wherein the first and second fins each comprise a portion of a bulk silicon substrate.

7. The integrated circuit structure of claim 1, wherein the first and second nanowires comprise silicon, and wherein the semiconductor material layer comprises silicon and germanium.

8. The integrated circuit structure of claim 1, wherein each of the first and second gate stacks comprise a high-k gate dielectric layer and a metal gate electrode.

9. An integrated circuit structure, comprising:

a first nanowire above a first fin, a channel region of the first nanowire electrically coupled to the first fin by a semiconductor material layer directly between the channel region of the first nanowire and the first fin but not between source or drain regions of the first nanowire and the first fin;

a first gate stack over the first nanowire;

a second nanowire above a second fin, a channel region of the second nanowire electrically isolated from the second fin;

a second gate stack over the second nanowire; and

a pair of epitaxial source or drain structures at first and second ends of the second nanowire.

10. The integrated circuit structure of claim 9, wherein the pair of epitaxial source or drain structures is a pair of non-discrete epitaxial source or drain structures.

11. The integrated circuit structure of claim 9, wherein the pair of epitaxial source or drain structures is a pair of discrete epitaxial source or drain structures.

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12. The integrated circuit structure of claim 9, wherein the first and second fins each comprise a portion of a bulk silicon substrate.

13. The integrated circuit structure of claim 9, wherein the first and second nanowires comprise silicon, and wherein the semiconductor material layer comprises silicon and germanium.

14. The integrated circuit structure of claim 9, wherein each of the first and second gate stacks comprise a high-k gate dielectric layer and a metal gate electrode.

15. A computing device, comprising:

a board; and

a component coupled to the board, the component including an integrated circuit structure, comprising:

a first nanowire above a first fin, a channel region of the first nanowire electrically coupled to the first fin by a semiconductor material layer directly between the first nanowire and the first fin;

a first gate stack over the first nanowire;

a second nanowire above a second fin, a channel region of the second nanowire electrically isolated from the second fin;

a second gate stack over the second nanowire; and

a pair of epitaxial source or drain structures at first and second ends of the second nanowire, wherein the pair of epitaxial source or drain structures is vertically separated from the second fin.

16. The computing device of claim 15, further comprising:

a memory chip coupled to the board.

17. The computing device of claim 15, further comprising:

a communication chip coupled to the board.

18. The computing device of claim 15, wherein the component is a packaged integrated circuit die.

19. The computing device of claim 15, wherein the component is selected from a group consisting of a processor, a communications chip, and a digital signal processor.

20. The computing device of claim 15, wherein the computing device is selected from a group consisting of a mobile phone, a laptop, a desk top computer, a server, and a set-top box.

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